

Banner appropriate to article type will appear here in typeset article

Transition over surface gaps in incompressible laminar boundary layers

Víctor Ballester Ribó¹, Jeffrey D. Crouch², Yongyun Hwang¹ and Spencer J. Sherwin¹

¹Department of Aeronautics, Imperial College London, South Kensington, London SW7 2AZ, UK

²The Boeing Company, P.O. Box 3707, Seattle, WA 98124-2207, USA

Corresponding author: Víctor Ballester Ribó, v.ballester-ribo24@imperial.ac.uk

(Received xx; revised xx; accepted xx)

The effect of surface gaps on the stability and transition of an incompressible laminar boundary layer is investigated by direct numerical simulation and linear stability analysis, for gap widths $w \in [0, 130]$, depths $d \in [0, 4]$ and Reynolds numbers $Re_{\delta^*} \in [800, 3000]$, based on the displacement thickness δ^* at the gap. For gaps that are narrow or shallow, the long-time two-dimensional dynamics settle onto an equilibrium. In this regime the gap acts solely as an amplifier, adding an increment Δn to the amplification factor n of Tollmien–Schlichting waves; transition remains downstream of the gap, but its location is shifted upstream of the smooth-wall reference point. Increasing the gap size destabilises the equilibrium, and the order in which stability is lost depends on the depth. Shallow gaps become unstable to two-dimensional perturbations first, and the flow settles onto a limit cycle that appears as a self-sustained oscillation around the gap. Deep gaps become unstable to purely three-dimensional perturbations first, but this instability saturates without provoking breakdown. Transition occurs at the gap precisely where the two-dimensional limit cycle is itself unstable to three-dimensional perturbations, a criterion that reproduces the bypass boundary measured experimentally. The analysis shows that both two- and three-dimensional instabilities are a necessary and sufficient condition for transition to occur at the gap.

1. Introduction

Aircraft surfaces that must sustain laminar flow are also manufactured surfaces. Panel joints, access doors and control-surface seams leave steps and gaps whose dimensions are set by manufacturing tolerances. Near a transport-aircraft wing's leading edge the displacement thickness is only $\delta^* \sim O(0.1 \text{ mm})$, so a millimetre gap spans several δ^* and can move transition far upstream. Tolerances are expensive to tighten. The question is not whether gaps can be removed but how large one may be before it matters.

Under the quiet free stream of flight, a boundary layer follows the classical route to turbulence: Tollmien–Schlichting (TS) waves are excited by receptivity, amplify convectively and break down at finite amplitude. In practice this sequence is reduced to the e^n method

(Smith & Gamberoni 1956; van Ingen 1956), which integrates the amplification factor $n(x)$ and declares transition where it reaches a threshold N calibrated to the disturbance environment (Crouch *et al.* 2015). Extending the method to an imperfect surface can be done in two ways. Either it increases the amplification, by the increment $\Delta n = n_{\text{gap}}(x) - n_{\text{flat}}(x)$, over the smooth-wall curve at fixed N , or it lowers the threshold, by the drop ΔN in n_{flat} , between the transition locations measured without the imperfection and with it (Crouch *et al.* 2006; Perraud *et al.* 2014; Crouch 2022). However, both methods rely on the same premise: the imperfection amplifies waves reaching it from upstream and creates none of its own.

Nothing guarantees that premise. Flight tests first identified a gap-width Reynolds number beyond which transition moves forward through (Nenni & Gluyas 1966). Laminar cavities were later sorted into open and closed regimes by whether the shear layer reattaches to the floor (Sinha *et al.* 1982; Zhang *et al.* 2005; Crook *et al.* 2013). ONERA studies (Olive & Blanchard 1982; Séraudie 2010; Gentili 2012) then varied the gap width w and depth d independently and mapped, in the (w, d) plane, the threshold beyond which transition occurs at the gap, and Beguet *et al.* (2017) found the depth negligible on ΔN for $0.2 < d/w < 0.7$. For deep gaps, $d/w > 5$, the compressible simulations of Zahn & Rist (2016) at $M = 0.6$ instead found a non-monotonic dependence of Δn on depth, which they attributed to an acoustic feedback whose resonance condition is set by the depth.

Only recently has the (w, d) plane been swept systematically in a single facility. The experiments of Crouch *et al.* (2022), at low Mach number ($M < 0.1$) and low free-stream turbulence, span $0 < w < 110$ and $0 < d < 11$ and show that, wherever TS waves still control transition, the data follow one of two limiting regimes selected by the aspect ratio. For $d/w \rightarrow 0$ the backward-facing step (BFS) result $\Delta N = 4.4d$ applies independently of w while for $d/w \rightarrow \infty$, $\Delta N \approx 0.1w$ applies independently of d . Beyond a boundary lying within $d \gtrsim 1.8$ and $w \gtrsim 19$, neither of these formulas holds: transition moves onto the gap itself and occurs at a markedly higher frequency, outside the TS band. These points were labelled *bypass* transition and excluded from the correlations.

This gap-located transition has been examined most directly by Medeiros and co-workers, all in compressible subsonic flow. Using global stability analysis at Mach numbers in the range $0.1 \leq M \leq 0.5$, Mathias & Medeiros (2019) attributed it to an instability of the shear layer spanning the gap, sustained by the edge-to-edge feedback loop of Rossiter (1964). Extending the range to $0.1 \leq M \leq 0.9$, Mathias & Medeiros (2021) showed that the thickness of the incoming boundary layer selects which Rossiter mode dominates, whereas the Mach number controls the acoustic energy transfer that closes the feedback loop. Thus, compressibility was found to be strongly destabilising even at low M . Finally, at $M = 0.5$, Aguirre *et al.* (2024) partitioned the (w, d) plane into three regions: one in which all modes are stable, one in which only three-dimensional centrifugal modes are unstable, and one in which two-dimensional Rossiter modes are unstable as well. The boundary of this last region lies close to the bypass boundary of Crouch *et al.* (2022). However, the nonlinear evolution of the three-dimensional modes, and how they interact with the two-dimensional ones, has received little attention.

Three questions are therefore left open. The first is where the amplifier description underlying both Δn and ΔN ceases to apply, that is, what distinguishes a gap that merely amplifies waves arriving from upstream from one that sustains an instability of its own. The second concerns gaps that remain amplifiers: what mechanism sets the amplification, and why gaps of different sizes amplify differently. The third concerns gaps that sustain their own instability, where transition is bypassed and moves onto the gap. Linear instability marks where the steady flow ceases to exist, not where transition occurs, and a saturated unstable mode may or may not break down. It thus remains to establish

what the unstable modes do once they saturate, at what stage the flow becomes three-dimensional, and whether the two- and three-dimensional instabilities act independently, compete or reinforce one another. This last question has no counterpart in the canonical cavity literature (Gharib & Roshko 1987; Rowley & Williams 2006; Brès & Colonius 2008; Citro *et al.* 2015), in which the cavity sits in an otherwise stable stream rather than in a boundary layer that is itself convectively unstable.

We address these questions using direct numerical simulation (DNS) and global and local stability analysis (Huerre & Monkewitz 1990; Chomaz 2005; Theofilis 2011) of incompressible flow over a spanwise-uniform rectangular gap, for $w \in [0, 130]$, $d \in [0, 4]$ and $Re \in [800, 3000]$. Narrow and shallow gaps settle onto an equilibrium and act purely as amplifiers, adding an increment Δn . The local analysis shows that Δn is set by a competition between two factors: the range of frequencies destabilised by the cavity shear layer, and the streamwise distance over which this destabilisation acts. Larger gaps destabilise the equilibrium, and the order in which stability is lost is set by the depth: shallow gaps lose stability to two-dimensional perturbations first and settle onto a limit cycle, whereas deep gaps lose it to purely three-dimensional perturbations, which saturate without breaking down. Transition occurs at the gap precisely where the two-dimensional limit cycle is itself unstable to three-dimensional perturbations, a criterion that reproduces the measured bypass boundary. Both instabilities are therefore necessary, and the onset of transition at the gap is set by a bifurcation of the flow rather than by an empirically-determined threshold.

The paper is organised as follows. Section 2 presents the governing equations and the numerical method. Section 3 computes the equilibrium base flows and their two- and three-dimensional linear stability. Section 4 examines the two-dimensional dynamics: the amplification increment Δn where the equilibrium is stable, and the limit cycles that emerge once the equilibrium loses stability. Section 5 follows the nonlinear three-dimensional dynamics and tests the limit cycles for stability to spanwise-periodic perturbations, which yields the transition scenarios at the gap. Section 6 summarises the results and discusses their implications.

2. Numerical setup

The incompressible Navier–Stokes equations are solved using a spectral/*hp*-element formulation. In nondimensional form, the governing equations are

$$\begin{aligned} \partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} &= -\nabla p + \frac{1}{Re} \Delta \mathbf{u}, \\ \nabla \cdot \mathbf{u} &= 0, \end{aligned} \tag{2.1}$$

where $\mathbf{u} = (u, v, w)^T$ denotes the dimensionless velocity field, p is the dimensionless pressure, and Re is the Reynolds number. Throughout the paper, $\mathbf{x} = (x, y, z)^T$ denotes the dimensionless spatial coordinates in three dimensions, with x , y and z denoting the streamwise, wall-normal and spanwise directions, respectively. The equations are nondimensionalised using the free-stream velocity u_∞ as the characteristic velocity and the displacement thickness δ^* as the characteristic length scale. The latter is evaluated at the upstream edge of the gap for the smooth-wall reference flow. Thus, $Re \equiv Re_{\delta^*} := u_\infty \delta^* / \nu$, where ν is the kinematic viscosity of the flow. The Reynolds number is varied in the range $Re \in [800, 3000]$. All the quantities in the paper are nondimensional unless otherwise stated.

All simulations are performed using the Nektar++ solver (Moxey *et al.* 2020), configured with a velocity-correction scheme and a continuous Galerkin projection (Serson *et al.*

2016; Karniadakis *et al.* 1991). Time integration is performed using a second-order IMEX scheme, with the viscous terms treated implicitly and the advection term treated explicitly. Spectral vanishing viscosity is added to stabilise the numerical scheme in regions containing sharp gradients (Kirby & Sherwin 2006).

The computational domain Ω is a rectangular cuboid, with a cavity of width w , depth d and spanwise-uniform geometry attached to the lower wall to model the surface gap (figure 1). The origin, $\mathbf{x} = \mathbf{0}$, is located at the upstream edge of the gap on the outermost spanwise plane. The domain dimensions are $h = 150$, $\ell_i = 100$ and $\ell_o = 1000$, which are chosen to ensure a fully developed upstream boundary layer and to minimise the influence of the upper and downstream boundaries on the region of interest. The spanwise length is set to $b = 4d$, which is sufficient to capture the relevant three-dimensional dynamics (see Section 3). A spanwise-uniform Blasius profile ($\mathbf{u} = \mathbf{u}_{BL}(y)$) is prescribed at the inlet and no-slip conditions ($\mathbf{u} = \mathbf{0}$) are imposed on the solid walls. At the upper boundary, a far-field condition ($\mathbf{u} = \mathbf{u}_\infty(x)$) is imposed. At the outlet, the outflow boundary condition of Dong *et al.* (2014) is used to minimise numerical instabilities when energetic vortices leave the computational domain. Periodic boundary conditions are imposed in the spanwise direction.

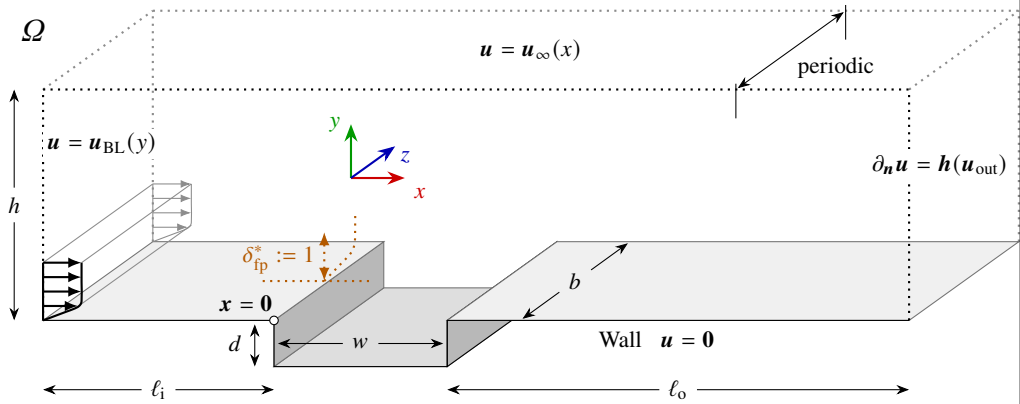


Figure 1. Computational domain used throughout the study. The gap is spanwise uniform, of width w and depth d , and the domain is periodic in the spanwise direction with spanwise length b . Boundary conditions are shown for the nonlinear simulations, where the function h in the outlet denotes Dong *et al.* (2014) outflow condition. The orange marker is the displacement thickness δ^* of the smooth-wall reference flow at the upstream edge of the gap, which sets the reference length. The dimensions are anisotropically rescaled for clearer visualisation of the gap geometry.

The domain is discretised using a Fourier expansion in the homogeneous spanwise direction, combined with a spectral/ hp element discretisation in each Fourier plane. Within a plane, a structured mesh of quadrilateral elements is used in the boundary-layer region and unstructured triangular elements in the far field. Between 4000 and 7000 elements are used in each simulation, depending on the gap geometry. On each element, a polynomial order of $p = 5$ and $p = 4$ is used to approximate the velocity and the pressure fields, respectively. This choice satisfies the inf-sup condition and prevents the numerical instabilities associated with an incompatible velocity-pressure discretisation (see Gatica 2014; Olshanskii *et al.* 2021). The polynomial basis is constructed from modified hierarchical Legendre polynomials of the corresponding dimensions (Karniadakis & Sherwin 2005). In the spanwise direction the solution is expanded using 48 homogeneous Fourier modes, so that the full three-dimensional discretisation amounts to approximately 4×10^6 degrees of freedom. The classical 3/2 padding dealiasing rule is incorporated in

the Fourier expansion and a spectral/*hp* dealiasing technique (Mengaldo *et al.* 2015) is used in the spectral/*hp* discretisation to prevent aliasing errors in the nonlinear terms. The resulting spatial resolution is sufficient to resolve the relevant flow features, as confirmed by grid-convergence studies.

For global linear stability analysis (LSA), the Navier–Stokes equations (2.1) are linearised around a base flow $(\mathbf{U}, P)^\top$, so that $\mathbf{u} = \mathbf{U} + \mathbf{u}'$ and $p = P + p'$, and the resulting perturbation equations for $(\mathbf{u}', p')^\top$ are solved. The base flows are obtained from nonlinear simulations of (2.1) and they can be either an equilibrium solution for classical LSA or a periodic orbit for Floquet analysis. When the base flow is globally unstable, the selective frequency damping method (Jordi *et al.* 2014) is used to suppress the unstable modes and obtain the corresponding unstable equilibrium solution. When the base flow is a stable periodic orbit, the periodic orbit is extracted through naive time integration. The linearised equations are then solved to determine the corresponding eigenvalues and eigenmodes. These can be compactly written as

$$\mathcal{M}\partial_t \mathbf{q}' = \mathcal{L}(\mathbf{U})\mathbf{q}',$$

$$\mathcal{M} = \begin{bmatrix} \mathbf{I} & 0 \\ 0 & 0 \end{bmatrix}, \quad \mathcal{L}(\mathbf{U}) = \begin{bmatrix} -\mathbf{U} \cdot \nabla - \mathbf{u}' \cdot \nabla \mathbf{U} + \frac{1}{Re} \Delta & -\nabla \\ \nabla \cdot & 0 \end{bmatrix}, \quad (2.2)$$

where $\mathbf{q}' = (\mathbf{u}', p')^\top$, $\mathbf{u}' = (u', v', w')^\top = \mathbf{u} - \mathbf{U}$ denotes the perturbation velocity field and $p' = p - P$ the perturbation pressure. For standard global LSA, the base flow is constant in time. For Floquet analysis, the base flow depends on time and it is T -periodic, $\mathbf{U}(t + T) = \mathbf{U}(t)$. For this study, in both cases the base flow is independent of the spanwise coordinate, so perturbations separate into spanwise Fourier modes that evolve independently. Combining this with Floquet's theorem, solutions of (2.2) are sought in the form

$$\mathbf{u}'(x, y, z, t) = \hat{\mathbf{u}}(x, y, t) e^{i\beta z} e^{\lambda t}, \quad \hat{\mathbf{u}}(x, y, t + T) = \hat{\mathbf{u}}(x, y, t), \quad (2.3)$$

where β is the spanwise wavenumber ($\beta = 0$ for two-dimensional perturbations) and λ is the complex Floquet exponent. For a spanwise-periodic domain of length b , the admissible wavenumbers are $\{\beta_n = 2\pi n/b : n = 0, \dots, 15\}$. When the base flow is steady, $\hat{\mathbf{u}}$ is independent of time and (2.3) reduces to the classical normal-mode ansatz. In both cases, the eigenvalue problem is solved with a matrix-free time-stepper approach (Barkley *et al.* 2008). Integrating the linearised equations (2.2) over a time interval τ defines the monodromy operator $\mathcal{A}(\tau)$, which maps an initial perturbation $\mathbf{q}'(0)$ onto $\mathbf{q}'(\tau)$. The eigenvalues μ of $\mathcal{A}(\tau)$ are related to the exponents by $\mu = e^{\lambda\tau}$, so that the eigenvalues of largest modulus correspond to the largest growth rates $\Re(\lambda)$. These leading eigenvalues and their eigenmodes are computed with an Arnoldi method that requires only repeated actions of $\mathcal{A}(\tau)$, that is, successive linearised simulations.

Local LSA complements the global analysis and is used mainly to interpret the amplification of two-dimensional TS waves across the gap. In this framework, the base flow is approximated as locally parallel at each streamwise location, $\mathbf{U} = (U(y), 0)$, and perturbations are sought as normal modes of the form

$$\mathbf{u}'(x, y, z, t) = \hat{\mathbf{u}}(y) e^{i(\alpha x + \beta z - \omega t)}, \quad (2.4)$$

where $\alpha = \alpha_r + i\alpha_i$ is the complex streamwise wavenumber, β the spanwise wavenumber and $\omega = \omega_r + i\omega_i$ the complex frequency. Since TS waves are two-dimensional, only $\beta = 0$ is considered, and substituting (2.4) into the linearised equations gives the Orr–Sommerfeld

problem (Schmid & Henningson 2001)

$$(D^2 - \alpha^2)^2 \hat{v} - iRe [(\alpha U - \omega)(D^2 - \alpha^2) - \alpha U''] \hat{v} = 0, \quad (2.5)$$

for the wall-normal velocity component $\hat{v}$, where $D = d/dy$. The problem is subject to $\hat{v}(0) = D\hat{v}(0) = 0$ and $\hat{v}, D\hat{v} \rightarrow 0$ as $y \rightarrow \infty$. Fixing Re , the pairs (α, ω) for which (2.5) admits a non-trivial solution form the zero set of a dispersion function $\mathcal{D}(\alpha, \omega; Re)$. The temporal and spatial formulations are two different real restrictions of this same zero set: the temporal one prescribes $\alpha = \alpha_r \in \mathbb{R}$ and solves for ω , whereas the spatial one prescribes $\omega = \omega_r \in \mathbb{R}$ and solves for α . In this work, only the spatial formulation has been used (see Section 4.1). Equation (2.5) is discretised by Chebyshev collocation on 81 Gauss–Lobatto points, with the domain truncated at $y_{\max} = 45$. The points are clustered near the wall by the algebraic mapping of Malik (1990). Following Bridges & Morris (1984), the auxiliary variable $\hat{q} = (D^2 - \alpha^2)\hat{v}$ is introduced to reduce it to a quadratic problem in α , which is then linearised into a generalised eigenvalue problem and solved with the QZ algorithm. Each streamwise station and frequency is solved independently, and the TS mode is selected from the full spectrum as the least damped eigenvalue whose phase speed lies in $0.15 < \omega_r/\alpha_r < 0.75$.

3. Global stability analysis

3.1. Base flows

For sufficiently small gap depths and widths, the flow asymptotically approaches an equilibrium state (i.e. laminar steady base flow), provided that the initial condition lies within its basin of attraction. The equilibrium flow topology in the boundary layer outside the gap remains close to a Blasius profile, while the flow topology inside the cavity changes strongly with the cavity aspect ratio w/d (figure 2).

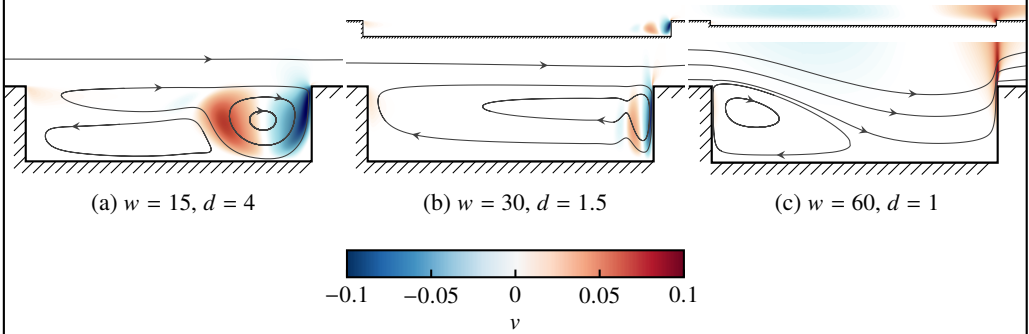


Figure 2. Equilibrium solutions (laminar steady base flow) for three representative gap geometries shown using wall-normal-velocity contours and streamlines. All geometries are rescaled to match the aspect ratio of the first case ($w/d = 15/4$) for visual comparison; the original unscaled domain profiles are shown above the last two panels.

For small w/d (figure 2a), a recirculation vortex forms within the cavity beneath the shear layer spanning the gap. This topology is characteristic of an open cavity, for which the separated shear layer remains detached from the cavity floor and the cavity interior is dominated by recirculating flow (Sinha *et al.* 1982; Crook *et al.* 2013).

Increasing the aspect ratio w/d (figure 2b) produces a more elongated, shallow recirculation region. The internal reverse flow is increasingly confined in the wall-normal direction,

while the flow still features a recirculation vortex in the vicinity of the downstream edge of the gap.

Increasing w/d further (figure 2c), the cavity is very shallow and the recirculating motion is confined predominantly to a region adjacent to the upstream edge of the gap. In contrast to the deep-cavity case, the outer shear layer exhibits strong streamwise curvature. The resulting structure approaches the closed-cavity topology, for which the separated shear layer reattaches to the cavity floor before leaving the cavity (Zhang *et al.* 2005).

In the limit $w/d \rightarrow \infty$, the two cavity steps become asymptotically separated in the streamwise direction, and the flow may be interpreted locally as a BFS flow in the vicinity of the upstream edge and a forward-facing step (FFS) flow near the downstream edge. The shear layer separated from the upstream step extends across the gap and becomes increasingly less affected by the downstream edge. The associated recirculation remains confined near the upstream step. Near the downstream edge, the shear layer hits the wall and is deflected back into the external flow, resulting in a flow topology similar to that of a FFS.

3.2. Stability of the base flows

To assess the stability of the equilibrium states, a biglobal stability analysis is performed for both spanwise-uniform ($\beta = 0$) and spanwise-varying ($\beta \neq 0$) perturbations. The stability of base flows with respect to spanwise-uniform (two-dimensional) disturbances is characterised by the growth rate

$$\sigma_{2D}(w, d) = \max \{ \Re(\lambda) : \lambda \in \text{spec}(\mathcal{L}_{w,d}^{\beta_0}) \}, \quad (3.1)$$

where $\mathcal{L}_{w,d}^{\beta_n}$ denotes the discretised linearised operator of (2.2) for the gap configuration (w, d) and spanwise wavenumber β_n , and $\text{spec}(\cdot)$ denotes the spectrum of the operator. For spanwise-varying perturbations, the leading eigenvalue is computed for the range of spanwise wavenumbers β_n described above, and the maximum growth rate over these wavenumbers is retained,

$$\sigma_{3D}(w, d) = \max_{n=1, \dots, 15} \max \{ \Re(\lambda) : \lambda \in \text{spec}(\mathcal{L}_{w,d}^{\beta_n}) \}. \quad (3.2)$$

For every gap considered and $Re = 1000$, the maximum in (3.2) is attained around β_9 or β_{10} , well inside the sampled range, and the growth rate at the largest wavenumber β_{15} is small compared with this maximum. The set $\{\beta_n\}_{n=0}^{15}$ therefore contains the most unstable wavenumber and is sufficient to capture the leading three-dimensional instability.

The onset of instability is determined by identifying the values of w and d for which σ_{2D} and σ_{3D} become positive. This procedure defines the two-dimensional and purely three-dimensional stability boundaries in the (w, d) plane. The frequency of the most unstable two-dimensional mode is given by

$$\omega_{2D} = \Im \left(\arg \max \{ \Re(\lambda) : \lambda \in \text{spec}(\mathcal{L}_{w,d}^{\beta_0}) \} \right), \quad (3.3)$$

and analogously for the frequency ω_{3D} of the three-dimensional modes. Figure 3 compares the resulting stability boundaries with the experimental bypass transition boundary reported by Crouch *et al.* (2022) at $Re = 1000$.

For deep gaps ($d \gtrsim 1.7$), the purely three-dimensional stability boundary lies at smaller gap widths than the two-dimensional boundary. This indicates that spanwise disturbances can destabilise the flow before the equilibrium becomes unstable to two-dimensional disturbances. For shallower gaps ($1.4 \lesssim d \lesssim 1.6$), the opposite behaviour is observed, with the two-dimensional instability occurring first. For deep gaps, the two-dimensional

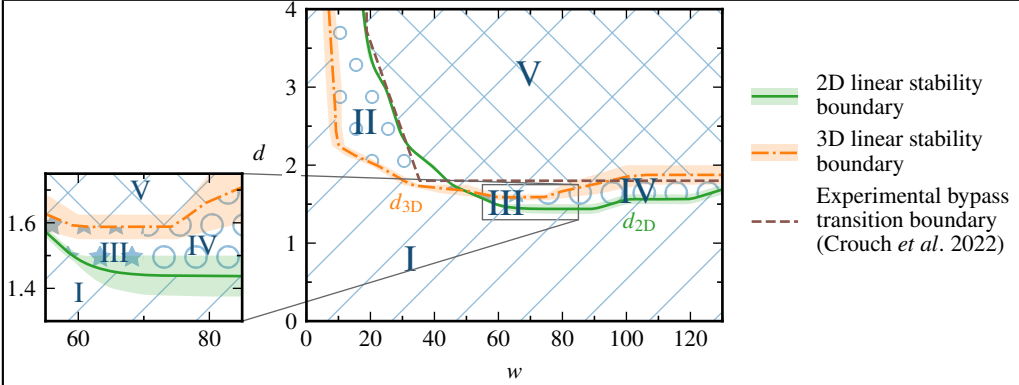


Figure 3. Two-dimensional (solid green line) and three-dimensional (dash-dotted orange line) linear stability boundaries in the (w, d) plane at $Re = 1000$. Shaded bands indicate the interpolation uncertainty associated with the discrete simulation grid. The dashed brown line shows the experimental boundary beyond which transition occurs at the gap location rather than through the downstream primary instability mechanism (Crouch *et al.* 2022). The different regions indicate the corresponding dynamical regimes discussed in the text.

stability boundary closely follows the experimental boundary separating transition through downstream TS waves from transition initiated at the gap location reported by Crouch *et al.* (2022). For shallow gaps, however, the three-dimensional stability boundary provides a closer agreement with the experimental boundary.

To make this comparison precise, note that within the domain considered, $(w, d) \in [0, 130] \times [0, 4]$, each stability boundary has at most one critical depth for a given gap width. The two-dimensional and three-dimensional boundaries can therefore be written as functions $d = d_{2D}(w)$ and $d = d_{3D}(w)$, respectively. Truncation at the largest depth considered here, $d = 4$, restricts their domains of definition to $[w_{2D}^c, 130]$ and $[w_{3D}^c, 130]$, where w_{2D}^c and w_{3D}^c are the widths at which each boundary attains that depth, $d_{2D}(w_{2D}^c) = d_{3D}(w_{3D}^c) = 4$. Since $w_{3D}^c < w_{2D}^c$ (figure 3), only the three-dimensional boundary lies within the domain for $w \in [w_{3D}^c, w_{2D}^c]$, whereas both do for $w \in [w_{2D}^c, 130]$. On the latter interval the two boundaries cross, and the experimental bypass transition boundary is closely described by their upper envelope,

$$w \mapsto \max\{d_{2D}(w), d_{3D}(w)\}, \quad w \in [w_{2D}^c, 130], \quad (3.4)$$

so that transition is set by the two-dimensional boundary for the deeper, narrower gaps and by the three-dimensional one for the shallower, wider gaps. Global flow instability, on the other hand, occurs after crossing the lower envelope

$$w \mapsto \begin{cases} d_{3D}(w), & w \in [w_{3D}^c, w_{2D}^c], \\ \min\{d_{2D}(w), d_{3D}(w)\}, & w \in [w_{2D}^c, 130]. \end{cases} \quad (3.5)$$

The close agreement between the experimental boundary and the stability results suggests that both two- and three-dimensional instabilities play a role in bypassing the primary transition mechanism of the boundary layer (see Section 5 for further discussion).

Together, the two envelopes partition the parameter space of figure 3 into regions of distinct linear stability. Below the lower envelope lies region I, in which equilibria is stable to both types of perturbation; above the upper envelope lies region V, in which equilibria is unstable to both. Between the two envelopes only one of the boundaries has been crossed, and which one depends on the gap width. For the narrower gaps, where $d_{3D} < d_{2D}$, the three-dimensional boundary is crossed first and the intermediate region II consists

of equilibria that are unstable to three-dimensional perturbations while remaining stable to two-dimensional ones. For the wider gaps, where $d_{2D} < d_{3D}$, the intermediate strip consists of equilibria that are unstable to two-dimensional perturbations while remaining stable to purely three-dimensional disturbances. This strip is split into regions III and IV, which share the same linear classification and are distinguished instead by their nonlinear dynamics, discussed in Section 4.

The two modes that define these boundaries differ markedly in spatial structure, in spatial extent and in time scale. The two-dimensional mode is uniform in the spanwise direction and corresponds to an oscillation of the cavity shear layer, whereas the purely three-dimensional mode has a non-zero spanwise wavenumber and takes the form of streamwise-elongated structures of alternating sign, reminiscent of the streaks observed in bypass transition of smooth-wall laminar boundary layers (Matsubara & Alfredsson 2001; Schmid & Henningson 2001). Figure 4 compares the two for the gap configuration $(w, d) = (19, 4)$ at $Re = 1000$.

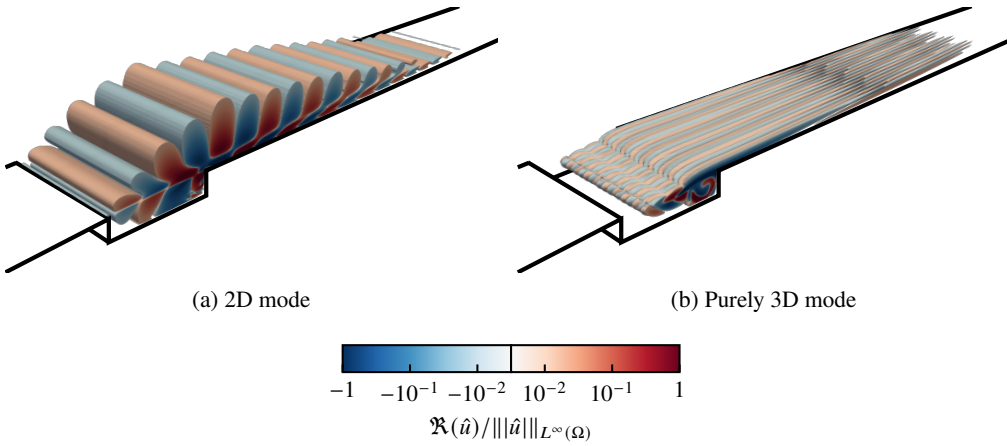


Figure 4. Real part of the streamwise component of the most unstable (a) two- and (b) purely-three-dimensional modes for the gap configuration $(w, d) = (19, 4)$ at $Re = 1000$. The modes are visualised using isosurfaces of $\Re(\hat{u})$ on a two-sided logarithmic scale.

The two-dimensional and purely three-dimensional modes are localised differently. The two-dimensional mode stays close to the gap, with a comparatively uniform amplitude near its downstream edge, whereas the three-dimensional mode extends far downstream in the form of long streamwise streaks. These streaks are much weaker than the mode structure inside the cavity. The two modes are also well separated in frequency: here $\omega_{3D} \approx 0.0256$, which is about an order of magnitude below $\omega_{2D} \approx 0.2457$, so the three-dimensional structures evolve on a much slower time scale than the shear-layer oscillation. A frequency separation of comparable size between the three-dimensional mode and the two-dimensional shear-layer mode is also reported for open cavities (Brès & Colonius 2008).

Both modes can be identified with the two families reported for compressible gaps by Mathias & Medeiros (2019, 2021) and Aguirre *et al.* (2024). The two-dimensional mode is an oscillation of the shear layer spanning the gap and corresponds to their Rossiter modes. In compressible flow, the Rossiter feedback loop is closed by acoustic waves propagating upstream from the downstream edge (Rossiter 1964). In the incompressible limit the pressure responds instantaneously to the impingement of the shear layer on that edge, so the feedback becomes purely hydrodynamic but the shear-layer oscillation persists.

The three-dimensional mode, concentrated in the recirculating region inside the cavity, corresponds to their centrifugal modes. It arises from a centrifugal instability of the closed streamlines of the recirculating vortex. This mechanism has already been identified in incompressible open cavities by Meseguer-Garrido *et al.* (2014), de Vicente *et al.* (2014) and Citro *et al.* (2015), and shown by Brès & Colonius (2008) to depend only weakly on the Mach number (for $M < 0.6$). The same physical mechanisms therefore appear to underlie both the present incompressible modes and the compressible modes of Medeiros and co-workers. One difference is that the partition of Aguirre *et al.* (2024) contains no region in which only the two-dimensional mode is unstable, whereas such a region (III and IV) is found here for wide, shallow gaps.

4. Two-dimensional dynamics

Building on the correspondence between the two-dimensional stability boundary and the experimental transition boundary, the nonlinear two-dimensional dynamics of flow over gapped surfaces are examined. A systematic nonlinear DNS parameter sweep is performed over gap depths d , gap widths w , and Reynolds numbers Re . The two-dimensional dynamical regimes identified at $Re = 1000$ are summarised in figure 5.

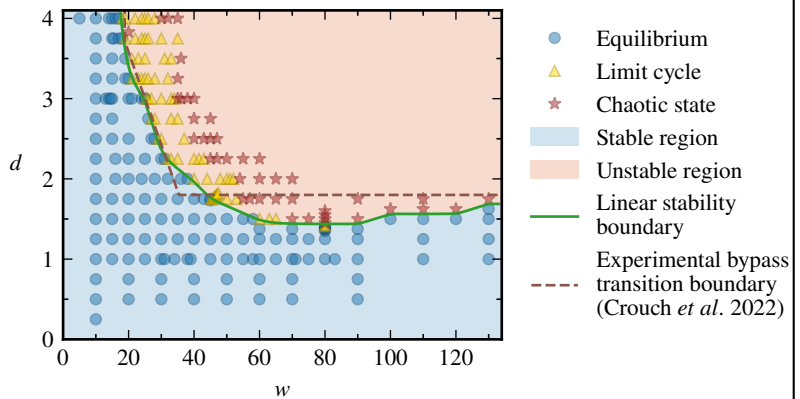


Figure 5. Limit set classification over the parameter space $0 \leq w \leq 130$ and $0 \leq d \leq 4$ at $Re = 1000$. Blue circles denote equilibrium states, yellow triangles denote limit cycles, and red stars denote chaotic states. The continuous green line represents the linear stability boundary separating the locally stable and unstable regimes, shaded in blue and red respectively. The dashed brown line represents the experimental boundary beyond which flow transition is observed at the gap location rather than through downstream primary instability mechanisms (Crouch *et al.* 2022).

As noted in Section 3, for sufficiently small gap depths and widths the flow asymptotically approaches an equilibrium state. For deeper gaps, increasing the width takes the flow through a sequence of bifurcations that ends in chaotic dynamics. The first of these is a supercritical Hopf bifurcation (solid green line in figure 5), at which the equilibrium gives way to a stable periodic orbit. Widening the gap further eventually destroys the periodic orbit in favour of chaotic motion. The full route to chaos has not been mapped out here because turbulence studied in this paper is three dimensional. However, several period-doubling bifurcations have been identified along the way (Section 4.2), which is a classical signature of such a route.

The shape of the stability boundary itself is informative, as it is non-monotonic in the gap width. Widening the gap is initially destabilising: the critical depth decreases until it reaches a minimum near $(w, d) \approx (75, 1.5)$, where the two steps interact most strongly.

Beyond that width the trend reverses and widening becomes stabilising, as the critical depth increases as w increases. As w increases further, the coupling between the two steps weakens and the critical depth approaches the value set by two isolated steps. DNS at $Re = 1000$, has shown that the onset of global instability is found at $d \in (2.75, 3.0)$ for the FFS and at $d \in (3, 3.25)$ for the BFS and very wide gaps are required to reach the asymptotic limit ($w \gg 250$).

4.1. TS waves amplification (2D stable regime)

TS waves are the dominant primary convective instability mechanism in two-dimensional laminar boundary layers. However, laminar boundary layers are globally stable to two-dimensional perturbations (Ehrenstein & Gallaire 2005). Therefore, a continuous source of noise is required to excite TS waves. To evaluate and corroborate the TS-wave amplification trends reported experimentally by Crouch *et al.* (2022), linearised simulations are performed about the equilibrium base states $\mathbf{U}$ obtained from DNS (blue circles in Figure 5; regions I and II in Figure 3). A body force $\mathbf{f} = (f_u, f_v)^T$ is added to the right-hand side of (2.2) to excite TS waves. The forcing $\mathbf{f}$ has a divergence-free Gaussian spatial envelope and is driven in time by zero-mean, unit-variance Gaussian white noise, X_t :

$$f_u(\mathbf{x}, t) = -(y - y_c) \exp(-\alpha|\mathbf{x} - \mathbf{x}_c|^2) X_t, \quad (4.1)$$

$$f_v(\mathbf{x}, t) = (x - x_c) \exp(-\alpha|\mathbf{x} - \mathbf{x}_c|^2) X_t, \quad (4.2)$$

where $\alpha = 100$, $\mathbf{x}_c = (x_c, y_c)^T = (-95, 0.5)^T$ and $|\cdot|$ denotes the Euclidean norm. The spatial envelope confines the forcing to a small circular region of radius approximately 0.5 centred at $\mathbf{x}_c$. Although ambient disturbances in experiments are neither strictly white nor Gaussian, this stochastic model provides a standard and reproducible representation of the external forcing.

The linearised system is integrated forward in time until the initial transient has decayed and a statistically stationary response is reached. The amplification is then quantified in two steps: first a streamwise-dependent scalar disturbance amplitude $A(x)$ is extracted from the solution, and then that amplitude is converted into an n -factor.

The amplitude at a streamwise position x is taken to be the wall-normal L^2 norm of the velocity perturbation,

$$A(x) := \|\mathbf{u}'\|_{L^2(\mathcal{Y}(x))} = \left(\int_{\mathcal{Y}(x)} (u'^2 + v'^2) dy \right)^{1/2}, \quad (4.3)$$

where $\mathcal{Y}(x)$ denotes the wall-normal section at x . Because the forcing is stochastic, the response fluctuates in time, and (4.3) is replaced by

$$A(x) := \left(\int_{\mathcal{Y}(x)} \langle u'^2 + v'^2 \rangle_{\mathcal{T}} dy \right)^{1/2}, \quad (4.4)$$

where $\langle \cdot \rangle_{\mathcal{T}}$ denotes temporal expectation over the stationary stochastic ensemble defined in the time interval $\mathcal{T}$.

Following the classical e^n method (Smith & Gamberoni 1956; van Ingen 1956), the growth of this amplitude relative to a reference position x_0 is measured by the local n -factor,

$$n_{x_0}(x) = \log \left(\frac{A(x)}{A(x_0)} \right). \quad (4.5)$$

The reference position x_0 is taken to be the location of minimum disturbance amplitude upstream of the gap. For brevity, the subscript is omitted hereafter, $n \equiv n_{x_0}$.

The amplification measured in this way can be compared with the prediction of local stability theory which, unlike the simulations, resolves the contribution of each individual frequency. Local LSA is therefore performed on the same equilibrium base flows at discrete streamwise positions, excluding the immediate vicinity of the gap corners, where the parallel-flow approximation is not valid. Because the forcing is white in time, the boundary layer is excited over a band $\mathcal{F}$ of real frequencies. Thus, the spatial formulation of the Orr–Sommerfeld problem is the relevant one: at each position x the wall-normal section $U(y; x)$ of the base flow is considered and (2.5) is solved for a prescribed real frequency $\omega_r \in \mathcal{F}$. The least stable mode $\alpha(x; \omega_r)$ and its associated eigenvector $\hat{\mathbf{u}}(y; x, \omega_r)$ are retained. The eigenvector is taken such that its L^2 norm over the wall-normal section $\mathcal{Y}(x)$ is one.

Since the base flow varies in x , the wavenumber does too, and the strictly parallel ansatz (2.4) must be replaced by its slowly varying counterpart. The perturbation is then written, in the manner of a WKBJ (Wentzel–Kramers–Brillouin–Jeffreys) approximation, as the superposition

$$\mathbf{u}'(x, y, t) = \int_{\mathcal{F}} \hat{\mathbf{u}}(y; x, \omega_r) e^{n(x; \omega_r)} \exp\left(i \int_{x_0}^x \alpha_r(\xi; \omega_r) d\xi\right) e^{-i\omega_r t} d\omega_r, \quad (4.6)$$

where

$$n(x; \omega_r) := \int_{x_0}^x -\alpha_i(\xi; \omega_r) d\xi, \quad (4.7)$$

so that $-\alpha_i$ is the local spatial growth rate and $n(x; \omega_r)$ the n -factor accumulated by that frequency alone between x_0 and x . Taking $\mathcal{T} = (0, \infty)$, the time average amplitude in (4.4) reduces to

$$A^2(x) = \int_{\mathcal{F}} e^{2n(x; \omega_r)} d\omega_r, \quad (4.8)$$

which yields the local version of the n -factor,

$$n(x) = \frac{1}{2} \log \left(\frac{1}{|\mathcal{F}|} \int_{\mathcal{F}} e^{2n(x; \omega_r)} d\omega_r \right), \quad (4.9)$$

where $|\mathcal{F}|$ is the measure of the frequency band. For the configurations considered, $\mathcal{F} = [0, 600]$, which is wider than the broadest unstable band of frequencies (see figure 7), so that the integral in (4.9) is dominated by unstable modes.

In figure 6 frequencies are shown using $F = 10^6 \omega_r \nu / u_\infty^2$ (Schmid & Henningson 2001), and streamwise positions through either x or the local Reynolds number based on the displacement thickness of the undisturbed Blasius boundary layer at that position,

$$Re_\ell(x) := \frac{u_\infty \delta^*(x)}{\nu} = Re \sqrt{1 + C^2 x / Re}, \quad (4.10)$$

with $C \approx 1.7208$. In figure 6a, the gap depth is fixed at $d = 1.5$ and the width is varied over $w \in [10, 50]$, while in figure 6b, the gap width is fixed at $w = 40$ and the depth is varied over $d \in [0.25, 1.75]$. In all cases, the first step of the gap starts at $x = 0$, and the shaded band spans across the length of the gap. The curves show a similar overall trend. Within the gap, n increases rapidly and reaches a maximum before the downstream edge. A similar peak in n was also observed experimentally by Crouch *et al.* (2022). Downstream of the gap, n decreases significantly before recovering a monotonic growth trend similar to that of the flat-plate reference, although with some streamwise variation in slope. Global and local LSA results agree closely. The local model reproduces the steep rise of n across

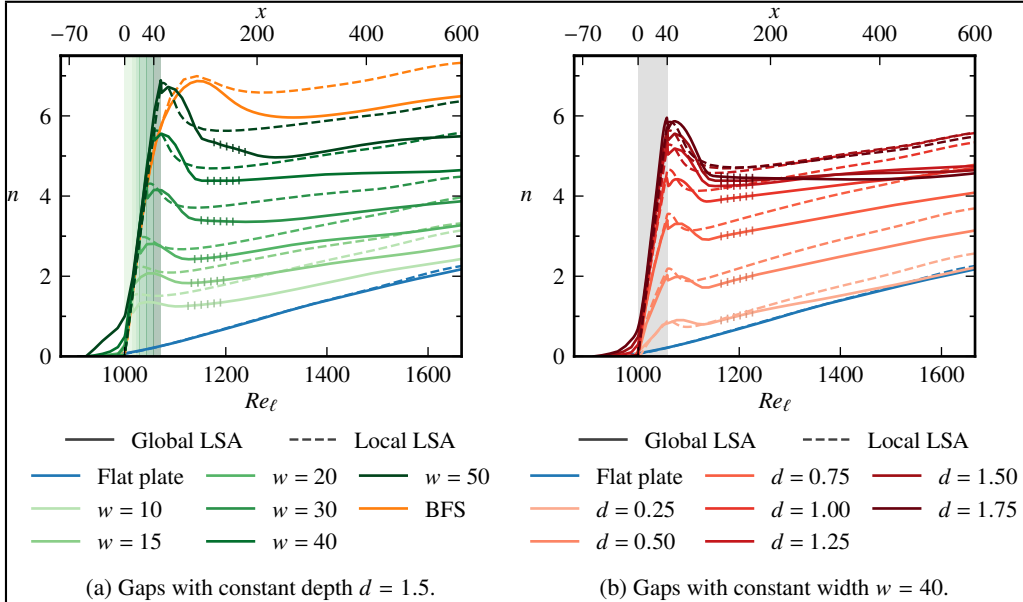


Figure 6. Streamwise evolution of the n -factor for different gap configurations at $Re = 1000$, from the linearised simulations (solid) and from the local analysis (dashed). Small markers indicate the streamwise locations used to compute the average Δn of (4.12). The flat-plate reference case is shown as a blue line and the BFS case as a orange line.

the gap, the maximum near the downstream edge, the drop that follows and the recovery at the flat-plate rate. The agreement is particularly good up to the peak of the n -factor curve. However, the local model does overestimate the amplification downstream. Downstream, far from the gap, the base flow resembles a Blasius profile, so the local growth rates, and with them the slopes of the local n -factor curves, become almost indistinguishable from those of the flat-plate case. The global results instead show a visibly smaller slope, and the discrepancy originates over the gap itself, where the flow is strongly non-parallel. There the low-frequency waves that the local analysis predicts to be convectively unstable downstream leave the gap having accumulated less amplification than the local model assigns them, which flattens the n -factor curves downstream of the gap.

For the constant-depth cases in figure 6a, the BFS case is also included for reference. The BFS produces the largest amplification, with the gap results approaching this value as the gap width increases, in agreement with the predictions of Crouch *et al.* (2022). Moreover, due to the absence of a second step, the abrupt drop observed in the gap cases is not present in the BFS curve. Instead, a more gradual decrease and smoother n -factor profile is observed. The differences in the subsequent recovery of the n curves after the initial drop are less clear. This is particularly evident in figure 6b, where the deeper gaps ($d \geq 1.25$) recover more slowly than the shallower gaps, resulting in intersections between some of the curves. To investigate the origin of these trends, figure 7 shows the neutral local stability curves $\{\mathcal{D}(\alpha, \omega) = 0 : \alpha, \omega \in \mathbb{R}\}$ of (2.5) using the same equilibrium base flows and gap configurations as in figure 6.

Several features of the local LSA help explain the observed behaviour of n in figure 6. Within the cavity, the unstable frequency range becomes broader. Since the system is forced with white noise, a wider range of frequencies is excited, resulting in a faster growth of the perturbations. This leads to the enhanced amplification observed in the n -factor curves.

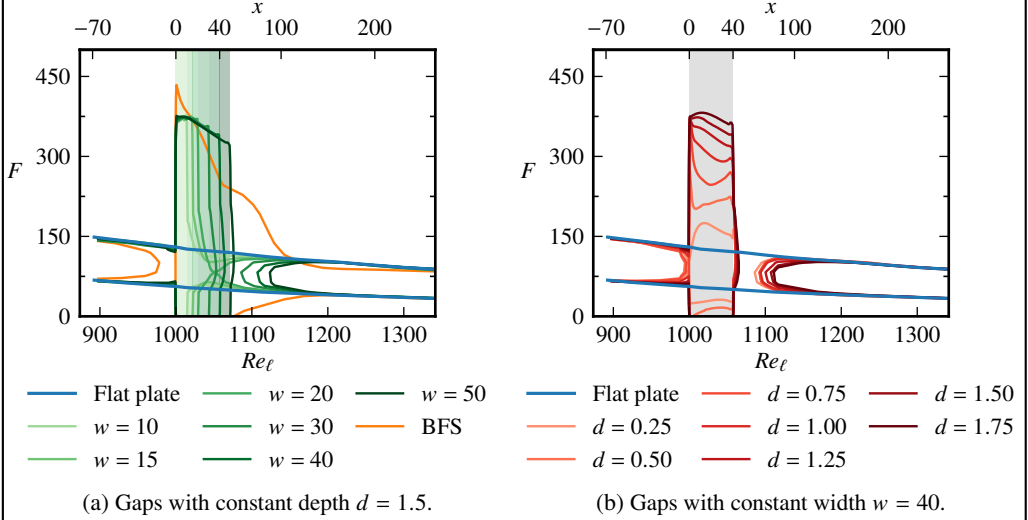


Figure 7. Neutral curves ($\alpha_i = \omega_i = 0$) of the Orr-Sommerfeld problem (2.5), for (a) constant gap depth $d = 1.5$ with varying width w , and (b) constant gap width $w = 40$ with varying depth d , compared with the flat-plate reference flow. The shaded band marks the location of the gap.

Downstream of small gaps, such as $w = 10$ and $w = 15$ in figure 7a, the unstable region gradually approaches the flat-plate neutral boundary, and the n curves recover towards the flat-plate reference. However, this recovery is local: the neutral curve and the local growth rate are properties of the base flow at a given position, whereas the perturbation amplitude in DNS is the accumulated result of the entire upstream history. Two effects therefore delay the recovery of the amplitude. First, the cavity alters the receptivity through non-parallel effects, so the frequency content entering the downstream boundary layer is weighted differently from that of the flat plate and that of the local LSA. Second, the problem is intrinsically advective, so this modified spectrum is transported downstream and amplified multiplicatively by the local growth rates, rather than being reset by them. The downstream evolution consequently does not necessarily need to follow the monotonic amplification of the flat-plate case, and even where the local stability boundaries collapse onto the flat-plate ones for $x \gg w$, the perturbation field from DNS does not immediately return to the flat-plate counterpart.

For wider gaps, the unstable region develops two distinct branches separated by a locally stable frequency range ($w \lesssim x \lesssim w + 20$ in figure 7b). This locally stable region corresponds to the temporary and abrupt reduction in n observed downstream of the wide gaps in figure 6. It also explains the slower recovery of the wider and deeper gaps shown in figure 6b as the stable frequency range broadens with increasing gap width and depth.

After understanding the shape of the n -factor curves, the next step is to quantify the effect of a surface imperfection on transition. The metric ΔN was introduced by Crouch *et al.* (2006) and applied to gaps by Crouch *et al.* (2022). This method is based on a change in the threshold value N at which transition is declared, and it is constructed from the measured shift of the transition location as follows. The n -factor curve of the smooth reference surface, $n_{\text{flat}}(x)$, is computed once and used throughout (the gap never enters the stability calculation). The transition location $x_{\text{flat}}^{\text{trans}}$ is measured in the wind tunnel for the smooth surface, giving the reference threshold $N_0 = n_{\text{flat}}(x_{\text{flat}}^{\text{trans}})$. For each gap configuration the transition location $x_{\text{gap}}^{\text{trans}}$ is then measured, and the threshold appropriate to that gap is the

projected value onto the n_{flat} curve, $N = n_{\text{flat}}(x_{\text{gap}}^{\text{trans}})$, so that $N = N_0 - \Delta N$ with

$$\Delta N := n_{\text{flat}}(x_{\text{flat}}^{\text{trans}}) - n_{\text{flat}}(x_{\text{gap}}^{\text{trans}}). \quad (4.11)$$

Since a gap shifts the transition location upstream, $x_{\text{gap}}^{\text{trans}} < x_{\text{flat}}^{\text{trans}}$ and hence $\Delta N > 0$. In the present work the smooth reference is the Blasius boundary layer. The gap correlations of Crouch *et al.* (2022) against which the results below are compared were instead obtained for a boundary layer developing under an airfoil pressure distribution.

In the present numerical framework, the transition location cannot be measured because the governing equations are linear. A metric Δn is therefore used to quantify how strongly the gap amplifies the disturbances arriving from upstream. Δn is defined as an averaged difference between the n -factor curve for a given gap configuration and the flat-plate reference flow, i.e.

$$\Delta n \equiv \Delta n_{\text{gap}} := \frac{1}{|X|} \int_X (n_{\text{gap}}(x) - n_{\text{flat}}(x)) dx, \quad (4.12)$$

where X is a streamwise interval located downstream of the gap and $|X|$ is its length. In the present study, X is taken to be 50 spatial units long and to begin downstream of the initial amplitude-recovery zone, where the direct effects of the gap and of the associated non-parallel flow have decayed. Small dashed markers on top of each amplification curve in figure 6 represent the streamwise locations in X used to compute the averaged Δn values. Figure 8 illustrates qualitatively the differences between the two metrics ΔN and Δn . Comparing the two therefore requires making some assumptions. In both, $n(x)$ is the envelope of the linear modes and N is the threshold value at which transition is declared. The two methods are equivalent provided that (i) transition is well correlated with a linear threshold amplitude, (ii) Δn captures the change in linear amplitude between the gap and the reference flow, and (iii) the measured ΔN corresponds to the same modes as those entering Δn . When these three conditions hold, raising the amplification curve by Δn and lowering the threshold by ΔN place transition at the same station, so that the two approaches yield the same result, $\Delta n = \Delta N$. Conditions (i) and (iii) fail together once the gap ceases to act as a mere amplifier and generates disturbances of its own, and the remainder of the paper is concerned with where that happens.

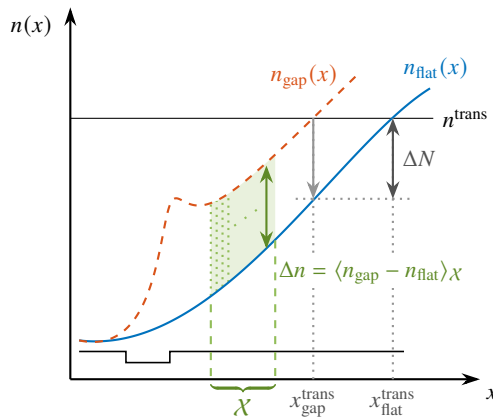


Figure 8. Schematic representation of the way the two metrics ΔN and Δn are computed. Subscripts *flat* and *gap* refer to the flat-plate reference flow and the flow over a generic gap, respectively. The superscript *trans* indicates the abstract transition location.

The metric Δn is evaluated for all parameter configurations that admit a stable steady base flow in the two-dimensional setting. To obtain a continuous representation over the whole stable parameter domain, the computed values are interpolated using Gaussian process regression with a Matérn kernel. Figure 9 compares the resulting numerical contours at $Re = 1000$ (solid black lines) with the experimental isolines $\Delta N = 1, 2, 3, 4, 5$ reported by Crouch *et al.* (2022) (dashed orange lines).

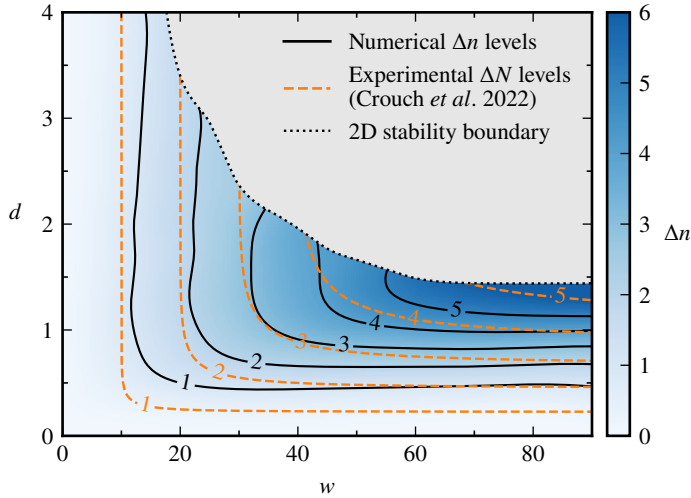


Figure 9. Contour map of Δn on the stable side of the parameter grid $(w, d) \in [0, 90] \times [0, 4]$ at $Re = 1000$. Black solid contours indicate the corresponding numerical predictions. Orange dashed contours indicate experimental levels from Crouch *et al.* (2022) for $\Delta N = 1, 2, 3, 4, 5$. The grey region is globally unstable and is excluded.

The agreement between the contours of Δn and the contours of ΔN is good. Both characteristic trends are reproduced: for narrow gaps ($w \lesssim 30$), the isolines are predominantly vertical, where the amplification is primarily determined by the gap width; for shallow gaps ($d \lesssim 1$), the isolines are predominantly horizontal, where the amplification is primarily determined by the gap depth. There is, however, a slight mismatch at higher Δn values, where the numerical predictions overestimate the experimental ΔN results. There are several possible reasons for this difference.

Although the TS-wave convective instability mechanism is two-dimensional, Crouch *et al.* (2022) conducted their experiments in a three-dimensional wind tunnel at low Mach number ($M < 0.1$). In contrast, the present simulations use a two-dimensional domain and solve the incompressible Navier–Stokes equations. This difference in dimensionality may contribute to the observed discrepancies, while the incompressibility assumption may also play a role. The forcing model is a further source of difference. White noise excites all frequencies with equal energy, whereas in the wind tunnel free-stream turbulence, acoustic noise and surface roughness excite some frequencies more strongly than others. Since the gap amplifies each frequency by a different amount, the two measures may be dominated by different frequencies. In the simulations, Δn is computed from the total amplitude $A(x)$, to which all forced frequencies contribute. In the experiment, ΔN is determined by the frequency that first reaches the transition threshold, and which frequency this is depends on the disturbance spectrum of the facility. Similarly, the disturbances are introduced here through a single localised source upstream of the gap, whereas in the experiment they enter the boundary layer along its whole length, including through the receptivity of free-stream disturbances at the gap edges, which the present model does not represent. In addition,

the choice of the averaging interval $\mathcal{X}$ in (4.12) introduces a small, but non-negligible effect on the resulting Δn values, particularly for gaps with large amplification, due to the streamwise variation of the n curves. Despite these sources of discrepancy, the computed Δn is in good qualitative agreement with the experimental ΔN .

Finally, the effect of Reynolds number is investigated. To study the effect of Reynolds number on the amplification of upstream disturbances, it is necessary to understand first how the linear stability boundary changes with Re . The dependence of the neutral stability curve on Reynolds number is shown in figure 10 for $Re \in [800, 3000]$. Increasing Re shifts the two-dimensional stability boundary towards smaller gap dimensions.

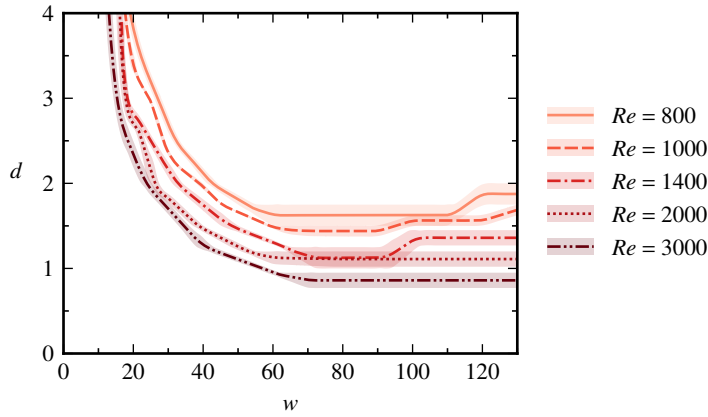


Figure 10. Linear stability boundaries in the (w, d) plane for $Re \in [800, 3000]$. Shaded bands indicate the uncertainty associated with interpolation between the discrete simulation points.

Figure 11 shows the Δn maps at $Re = 800$ and $Re = 3000$. Two effects act at the same time. First, the part of the parameter space that admits a stable steady base flow shrinks considerably as Re increases. Second, inside the stable region the shape of the Δn field is unchanged: the contours remain nearly vertical for narrow gaps and nearly horizontal for shallow gaps, similar to the trends observed at $Re = 1000$.

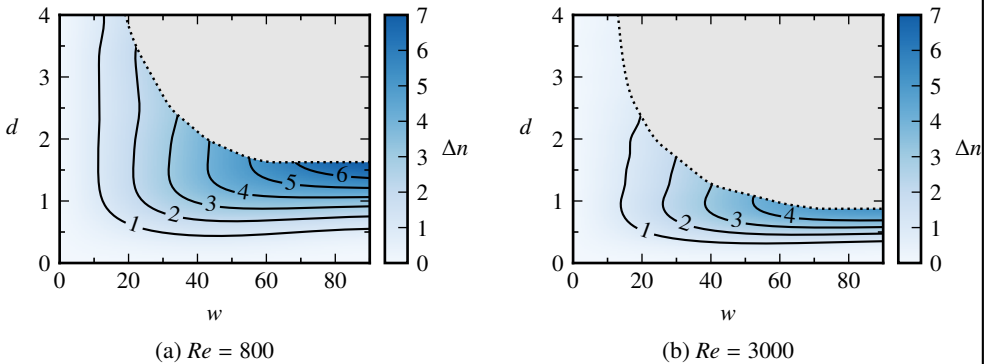


Figure 11. Contour maps of numerical predictions of Δn over the parameter space $(w, d) \in [0, 90] \times [0, 4]$ for (a) $Re = 800$ and (b) $Re = 3000$. The dotted line is the two-dimensional stability boundary; the grey region above it is globally unstable and is excluded.

Comparing both plots in figure 11 with figure 9 shows that the two limiting regimes respond to the Reynolds number in opposite ways: for shallow gaps Δn increases with

Re , whereas for narrow gaps it decreases. The Δn maps, however, measure only the total additional amplification supplied by a gap; they indicate neither where along the plate this amplification is acquired nor which frequencies carry it. To understand this behaviour better, and relying on the good agreement between global and local LSA, local analysis is therefore repeated for the two opposite corners of the parameter space that remain globally stable at every Reynolds number considered: a narrow deep gap, $(w, d) = (12, 4)$, and a wide shallow gap, $(w, d) = (90, 0.5)$.

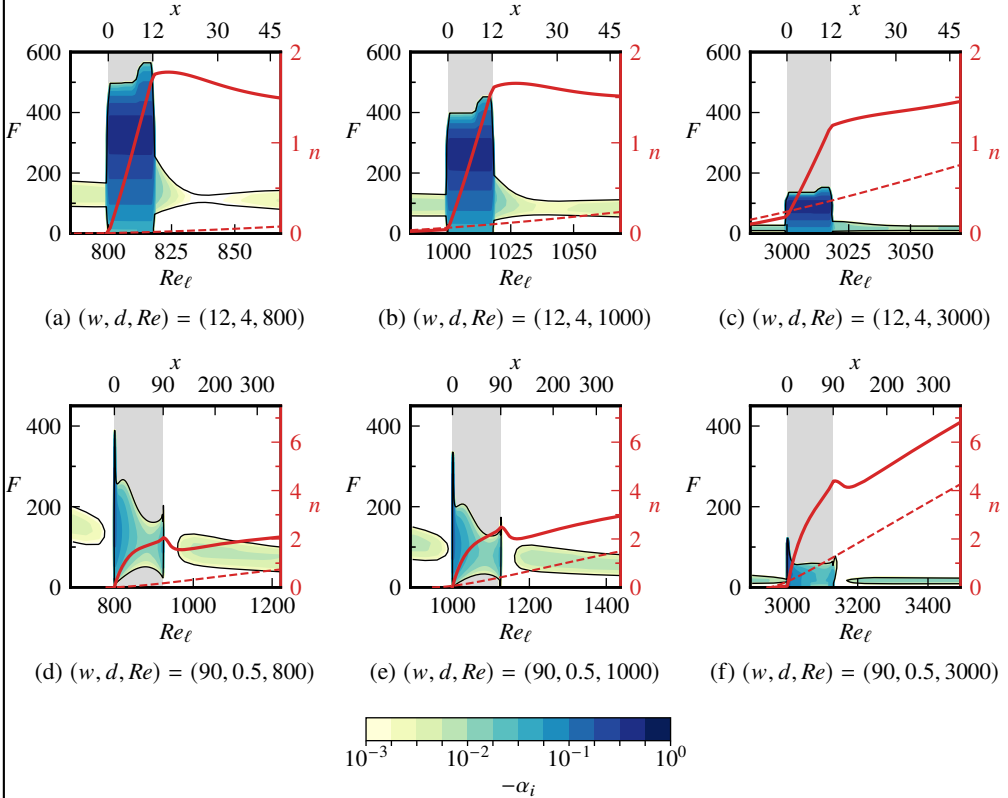


Figure 12. Local stability diagram and the n -factor it integrates to, for a narrow deep gap ($w = 12$, $d = 4$; top row) and a wide shallow gap ($w = 90$, $d = 0.5$; bottom row) at $Re = 800$ (left column), 1000 (middle column) and 3000 (right column). In each panel the bottom axis is the local Reynolds number Re_ℓ based on the local displacement thickness of the undisturbed Blasius boundary layer and the top axis is the streamwise position x measured from the upstream gap edge. The left axis is the frequency $F = 10^6 \omega_r \nu / u_\infty^2$ and the right axis the n -factor. The solid black line is the neutral curve $\alpha_i = 0$ and the filled contours inside it show the growth rate contour map. The solid red line is the n -factor (4.9) of the synthesised perturbation, and the dashed red line is its flat-plate counterpart. The shaded bands mark the location of the gaps, $0 \leq x \leq w$.

Figure 12 collects, for each of the two geometries and each Reynolds number, the neutral curve $\alpha_i = 0$, the growth rate $-\alpha_i > 0$ inside it and the n -factor (4.9) that the growth rates integrate to, together with its flat-plate counterpart. The shaded band marks the gap.

In all six panels the neutral curves have qualitatively the same shape as those of figure 7. Upstream of the gap the unstable region resembles that of the flat plate; over the gap it opens abruptly into a much broader band of frequencies, and within one or two gap widths downstream it has contracted back onto the flat-plate neutral curve. For the wide-gap case a locally stable region appears immediately upstream and downstream of the gap. Essentially all of the excess amplification is therefore produced inside the shaded band. Since n is the

streamwise integral of the growth rate over the amplified band, three quantities set it: the width of that band, the magnitude of $-\alpha_i$ within it, and the distance over which it acts. Increasing Re narrows the band but strengthens the shear layer, and which of the two prevails is decided by the length of the gap.

For the narrow deep gap (figures 12a–12c) the band broadens most at the lowest Reynolds number and narrows monotonically as Re increases. The shear layer is marginally more unstable at $Re = 3000$, but the gap is only 12 units long, so the higher growth rates act over too short a distance to offset the loss of bandwidth. The two effects do not compensate, and Δn decreases monotonically from figure 12a to figure 12c. For the wide shallow gap (figures 12d–12f) the balance is reversed. The band is again wider at $Re = 800$ than at $Re = 3000$, but the growth-rate isolines there are lower, and the gap extends over 90 units, which is a sufficient distance for the stronger shear layer of the higher Reynolds numbers to accumulate amplification. Here Δn increases monotonically from figure 12d to figure 12f. The local analysis therefore reproduces the opposite Reynolds-number trends of figure 11: narrow gaps amplify most at low Re , while wide gaps amplify most at high Re .

4.2. Limit cycles (2D unstable regime)

As the gap width or depth is increased the flow undergoes a supercritical Hopf bifurcation (solid green line in figure 5). The supercritical nature of the Hopf bifurcation is confirmed by the crossing of a single complex-conjugate pair of eigenvalues through the imaginary axis, while the remaining eigenvalues retain negative real parts. The expected normal-form scaling is further verified by tracking the amplitude of the emerging limit cycle along three continuation paths that cross the Hopf boundary in the (w, d) plane, shown in the inset of figure 13: γ_1 crosses it by varying the gap width at fixed depth, γ_2 along an oblique direction, and γ_3 by varying the depth at fixed width. Defining $\mu := \pm\sqrt{(w/w_c - 1)^2 + (d/d_c - 1)^2}$ as the relative distance from the critical point (w_c, d_c) where each path crosses the Hopf boundary, positive on the unstable side, the normal form for a supercritical Hopf bifurcation then predicts

$$A(\mu) \propto \sqrt{\mu}, \quad (4.13)$$

where A is an appropriate measure of the limit cycle amplitude. Figure 13 shows the squared L^2 norm of $|\mathbf{u}'|$ over the whole domain Ω along γ_1 and γ_2 . In both cases A^2 depends approximately linearly on μ .

The remaining path, γ_3 , crosses the Hopf boundary at fixed width $w = 80$ by increasing the depth through $d \approx 1.4$. A single complex-conjugate pair of eigenvalues crosses the imaginary axis here as well, yet the squared global norm does not grow linearly with μ . This is not a failure of the normal form but a consequence of measuring the amplitude with a global norm. For gaps this wide the limit cycle frequency has fallen to the upper edge of the band of frequencies that become convectively unstable somewhere downstream of the gap (see figure 14), so the oscillation at the downstream gap edge acts as a sustained harmonic source for the boundary layer behind it. The global L^2 norm then mixes two contributions: the oscillation localised at the gap edge, whose amplitude does scale as $\sqrt{\mu}$, and the downstream wave, whose amplitude is set by the integrated convective growth rate, and hence by the domain length and by how far below the upper branch of the local neutral stability curves the frequency lies, rather than by the distance from the bifurcation point. Restricting the norm to the region $\Omega' = [w - 10, w] \times [-d/2, 0.5]$, which encloses the end of the shear layer up to the downstream gap edge, isolates the first contribution. Measured in Ω' , the squared amplitude along γ_3 grows linearly with μ close to the bifurcation (figure 13), as expected from the normal form.

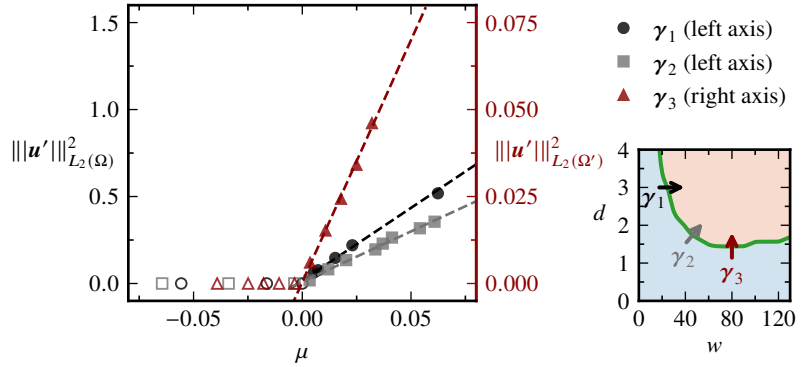


Figure 13. Amplitude evolution of the limit cycle for different crossing locations of the Hopf bifurcation in the (w, d) parameter plane. The three continuation paths are indicated in the small plot on the right. The abscissa μ is a measure of the distance from the bifurcation point along each path, so that the bifurcation lies at $\mu = 0$ on all three paths and $\mu > 0$ on the unstable side. Paths γ_1 and γ_2 (black and grey, left axis) are measured by the squared L^2 norm of $|\mathbf{u}'|$ over the whole domain Ω . Path γ_3 (dark red, right axis) is measured over $\Omega' = [w - 10, w] \times [-d/2, 0.5]$. Open symbols mark the stable side of the bifurcation; closed symbols mark the unstable side, and the lines are linear fits on the unstable side.

The fundamental frequency ω of the limit cycles varies with Reynolds number and cavity geometry, as summarised in figure 14. For narrow gaps ($w \lesssim 60$), the frequencies form three distinct modal branches associated with the cavity geometry and recirculation dynamics. As the gap width increases at fixed depth, the dominant mode of the narrower cavity loses stability and a different periodic state emerges, with a frequency more representative of the larger cavity. The weaker interactions between branches observed across different depths indicate a secondary dependence on the gap geometry. For wide gaps ($w \gtrsim 65$), the limit cycle frequencies approach the characteristic frequency range of convectively unstable TS waves at the corresponding Reynolds number. The associated modes have longer streamwise scales and lower frequencies.

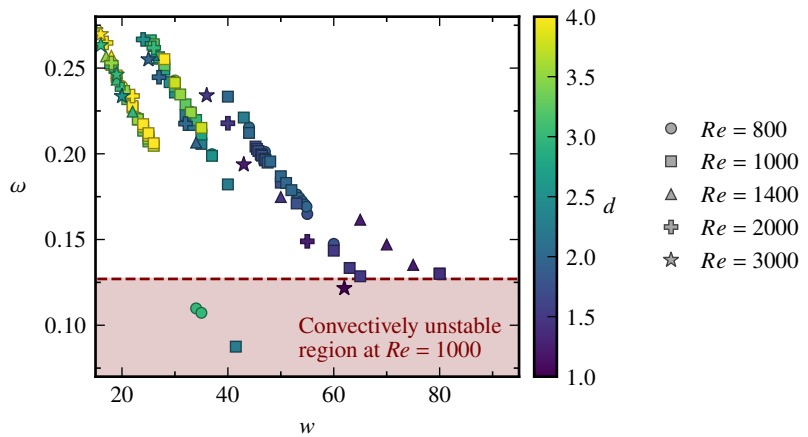


Figure 14. Limit cycle frequencies as a function of gap width w for different gap depths and Reynolds numbers. Marker shading indicates gap depth, with darker markers corresponding to shallower gaps and lighter markers corresponding to deeper gaps, and marker shape denotes Reynolds number. The red shaded region indicates the approximately constant frequency range of convectively unstable TS waves in the neighbourhood of the gap at $Re = 1000$.

Several isolated points around $w \approx 40$ and $\omega \approx 0.1$ in figure 14 correspond to states after a period-doubling bifurcation, suggesting a route from periodic to chaotic dynamics. Figure 15 illustrates one such bifurcation for two gap configurations at $Re = 800$. Increasing the gap width from $w = 33$ to 33.5 at $d = 3$ results in a limit cycle with approximately twice the period.

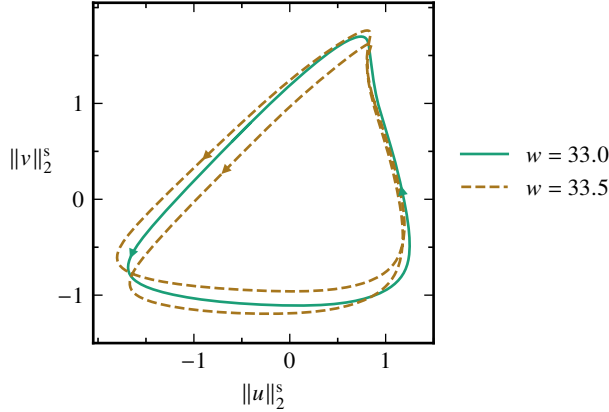


Figure 15. Periodic orbits for the gap configurations $(w, d) = (33, 3)$ and $(w, d) = (33.5, 3)$ at $Re = 800$ projected onto the two-dimensional space $(\|u\|_2^s, \|v\|_2^s)$. Here $\|\cdot\|_2^s := (\|\cdot\|_2 - \overline{\|\cdot\|_2})/\sigma(\|\cdot\|_2)$ denotes the standardised L^2 norm over the computational domain, with σ denoting the standard deviation of the corresponding time series.

5. Three-dimensional dynamics and transition

Even though the primary convective instability of the boundary layer is two-dimensional, the global stability analysis of Section 3 has shown that a purely three-dimensional mode can become unstable before the two-dimensional one when the gap is sufficiently deep (figure 3). This section examines the nonlinear three-dimensional dynamics in order to identify the transition scenarios that a gap can produce. Three cases are distinguished, according to which instability is active. If only the three-dimensional mode is unstable, the perturbation saturates and the downstream boundary layer remains laminar. If the equilibrium has also lost stability to two-dimensional disturbances, as happens for deep gaps, the two instabilities interact and transition is triggered at the gap itself. If instead the two-dimensional instability sets in first, as happens for wide and shallow gaps, three-dimensionality develops only further downstream. The three cases are treated in turn below.

First, region II of figure 3 is considered, where the purely three-dimensional mode has a positive growth rate while all two-dimensional modes are still stable. Nonlinear DNS initialised along this mode shows the expected exponential growth, until the perturbation saturates at a finite amplitude and the flow settles onto a low-variance time-varying three-dimensional state without breaking down. Figure 16 shows where that saturated perturbation resides for a gap with width $w = 24$ and depth $d = 2$ at $Re = 1000$ using the square root of a time- and spanwise-averaged energy of the perturbation velocity,

$$\eta := \sqrt{\langle |\mathbf{u}'|^2 \rangle_{\mathcal{Z} \times \mathcal{T}}} = \left(\frac{1}{b} \int_{\mathcal{Z}} \langle u'^2 + v'^2 + w'^2 \rangle_{\mathcal{T}} dz \right)^{1/2}, \quad (5.1)$$

where $\mathcal{Z} = [0, b]$ is the spanwise domain and $\langle \cdot \rangle_{\mathcal{T}}$ denotes temporal expectation over the time interval $\mathcal{T}$. This quantity is a measure of the average magnitude of the three-dimensional perturbation velocity in a spanwise 2D slice of the domain.

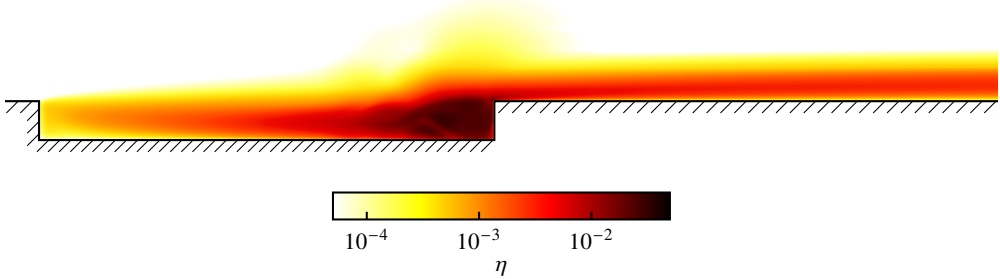


Figure 16. Distribution of the quantity η (5.1) in a region surrounding the gap of dimensions $w = 24$, $d = 2$ at $Re = 1000$.

Figure 16 shows where η resides. It is concentrated inside the cavity and around the downstream edge, where it reaches magnitudes comparable to that of the base flow, and only a weak residual extends into the rest of the gap and the boundary layer behind the gap, where η is small even by comparison with the base flow. Note that this distribution is qualitatively similar to that of the three-dimensional eigenmode shown in figure 4b. Downstream of the gap the three-dimensional perturbation is therefore too weak to excite the boundary layer and provoke transition on its own. However, the three-dimensional structures inside the cavity act as a sustained low-amplitude source of low-frequency noise for the boundary layer behind it, and thereby provide a continuous excitation of the TS waves downstream. These TS waves amplify as they are convected downstream and become visible only near the downstream boundary of the domain (not shown for brevity).

The dynamics are qualitatively different once the equilibrium becomes unstable to two-dimensional disturbances. Widening the gap beyond the two-dimensional stability boundary produces a chaotic state that develops predominantly near the downstream edge of the gap (figure 17). This behaviour corresponds to region V of figure 3.

Figure 17 illustrates the transition sequence for the deep-gap configuration $(w, d) = (19, 4)$ at $Re = 1000$. The flow is initialised with a small perturbation in the direction of the most unstable two-dimensional mode (figure 17a). It first evolves towards the corresponding periodic orbit, which is unstable to three-dimensional perturbations. As the instability develops, three-dimensional structures appear both within the cavity and downstream of the gap (figure 17b). The flow, then, subsequently departs from the two-dimensional periodic orbit and develops a chaotic state in the downstream boundary layer (figure 17c). At the later stages, the downstream flow exhibits increasingly complex three-dimensional structures leading to fully developed turbulence (figure 17d). Nevertheless, in the immediate vicinity of the downstream gap edge, the characteristic frequency of the underlying two-dimensional oscillation remains present. This behaviour is consistent with the strong non-TS-wave spectral peak reported experimentally by Crouch *et al.* (2022).

A qualitatively similar end state is reached from other initial conditions. Because the three-dimensional mode grows slightly faster than the two-dimensional one at this configuration in the linear regime, it is the first structure to emerge from a generic small disturbance, and the simulation has therefore also been initialised along the most unstable three-dimensional eigenmode. The disturbance then follows the predicted linear growth until nonlinear effects take over, at which point the flow leaves the eigenmode direction, the two-dimensional mode grows, and the flow again ends in a chaotic state similar to that

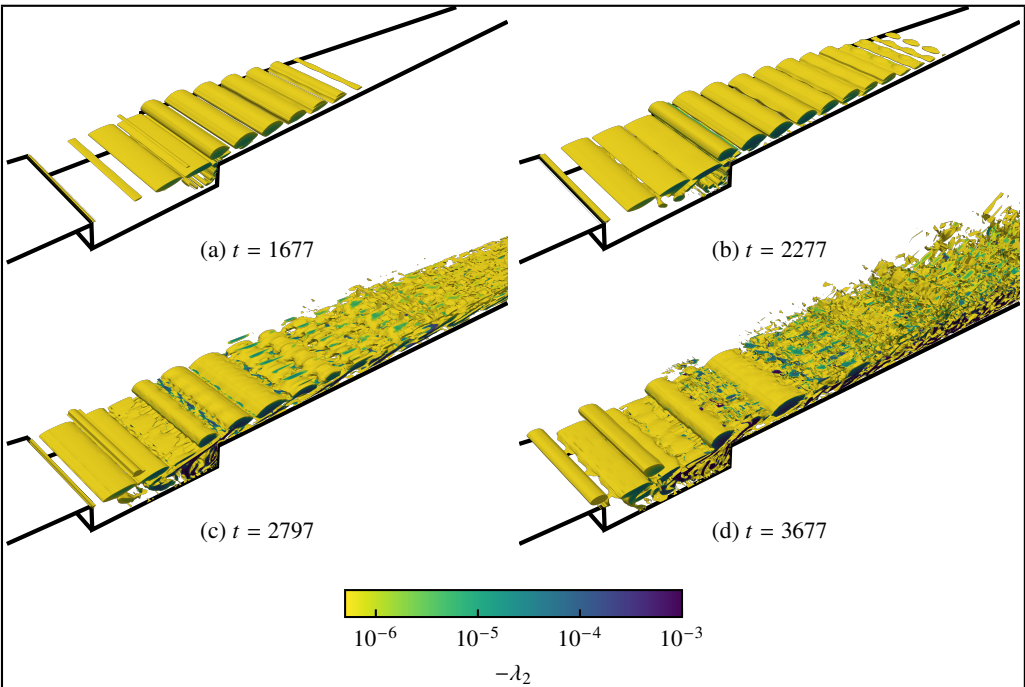


Figure 17. Transition sequence for $w = 19$ and $d = 4$ at $Re = 1000$. Vortical structures are visualised using λ_2 -criterion isosurfaces. The simulation time corresponding to each state is indicated next to each visualisation.

of figure 17d. Which mode is excited first therefore changes the route slightly, but not the final state.

Both routes can be interpreted in terms of the triadic interactions produced by the quadratic nonlinearity of the Navier–Stokes equations, through which two Fourier components with spanwise wavenumbers and frequencies (β_1, ω_1) and (β_2, ω_2) force a third at $(\beta_1 \pm \beta_2, \omega_1 \pm \omega_2)$. Interacting with itself, the three-dimensional mode $(\pm\beta, \omega_{3D})$ forces the harmonics $(2\beta, 2\omega_{3D})$ and $(2\beta, 0)$, and the spanwise-uniform components $(0, 0)$ and $(0, 2\omega_{3D})$. The latter components correspond to a distortion of the mean flow and a slow two-dimensional oscillation, respectively. Since $2\omega_{3D} \approx 0.05$ is far from $\omega_{2D} \approx 0.25$ and also because $\omega_{2D}/\omega_{3D} \approx 9.6$, this self-interaction cannot force the two-dimensional mode resonantly. The two-dimensional mode is instead linearly unstable on its own at this configuration and grows from the small spanwise-uniform content already present in the flow. Conversely, once the two-dimensional oscillation $(0, \omega_{2D})$ is established, its interaction with the three-dimensional perturbation transfers energy to the sidebands $(\beta, \omega_{3D} \pm k\omega_{2D})$, $k \in \mathbb{N}$, so that the spanwise structure is now modulated at the frequency of the shear-layer oscillation and its harmonics. Repeated interactions among these components and the harmonics of both modes populate a broadband spectrum in both β and ω , which corresponds to the chaotic state observed downstream of the gap.

This result, together with the saturated region-II case discussed above, shows that the three-dimensional instability alone does not cause breakdown. On its own, the three-dimensional mode transfers energy only to its harmonics and to slow spanwise-uniform components, and it saturates. Transition at the gap requires both instabilities: the three-dimensional mode provides the spanwise structure, and the interaction of this structure with the high-frequency two-dimensional oscillation spreads energy across the spectrum and drives the flow to turbulence. To better understand this interaction, a Floquet analysis

has been performed around the two-dimensional unstable limit cycle. The leading Floquet mode obtained in this way is shown in figure ?? for $(w, d) = (19, 4)$ at $Re = 1000$.

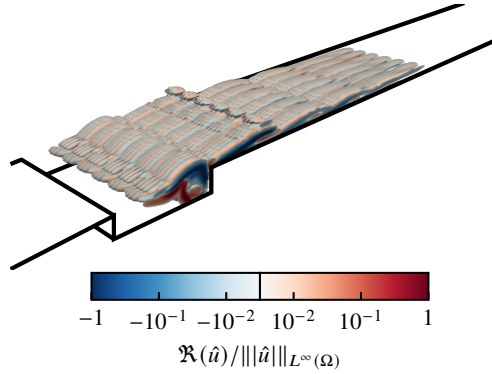


Figure 18. Most unstable floquet mode for the gap configuration with $w = 19$ and $d = 4$ at $Re = 1000$. The mode is visualised using isosurfaces of $\Re(\hat{u})$ on a two-sided logarithmic scale.

The shape of the mode is similar to that of the three-dimensional eigenmode, with a strong concentration of the perturbation inside the cavity and around the downstream gap edge. Weak long streamwise streaks extend into the downstream boundary layer. However, contrary to the streaks of the three-dimensional eigenmode in figure 4b, the streaks of the Floquet mode are modulated by the underlying two-dimensional oscillation.

The third scenario occupies regions III and IV of figure 3, in which the two-dimensional instability sets in before the three-dimensional one and the resulting limit cycle remains stable to three-dimensional perturbations. The subsequent evolution of the flow is therefore governed not by a secondary three-dimensional instability, but by where the frequency of the two-dimensional oscillation falls relative to the band of convectively unstable TS waves, which for shallow gaps lies close to the upper edge of that band (figure 14). This is what distinguishes the two regions. In region III the frequency lies above the upper branch, so the oscillation is damped as it is advected downstream and the boundary layer remains laminar. In region IV the frequency lies within the band, so the oscillation is amplified downstream of the gap and transition follows through the primary pathway of TS-wave growth.

The passage from region III to region IV is gradual, in contrast to the abrupt crossings from region I to region II or from region II to region V. Figure 19 illustrates the underlying mechanism. Deep within region III the oscillation is fully localised at the downstream edge of the gap and decays rapidly behind it. As the gap widens, the structure extends farther downstream and its decay becomes progressively slower, until it reaches far enough for the convectively unstable band to amplify it, at which point transition occurs.

Figure 19 shows this directly, giving the wall-normal L^2 norm of the perturbation velocity against distance from the downstream gap edge for different gap widths at fixed $d = 1.5$ and $Re = 1000$. The oscillation is localised at the gap edge in every case; what changes with w is its downstream evolution. For $w = 60$ and 63 the frequency lies above the upper branch at the gap, the excited wave is damped everywhere downstream, and the norm decays to negligible values within a few hundred spatial units. At $w = 65$ the frequency lies inside the unstable band over part of the domain only: the norm grows to a maximum near $x - w \approx 400$ and decays once the wave leaves that region. For $w \geq 70$ the wave remains amplified over the whole domain and the norm grows monotonically to the end

of it. In this latter case, the global and convective mechanisms reinforcing one another, leaving the downstream boundary layer fully destabilised.

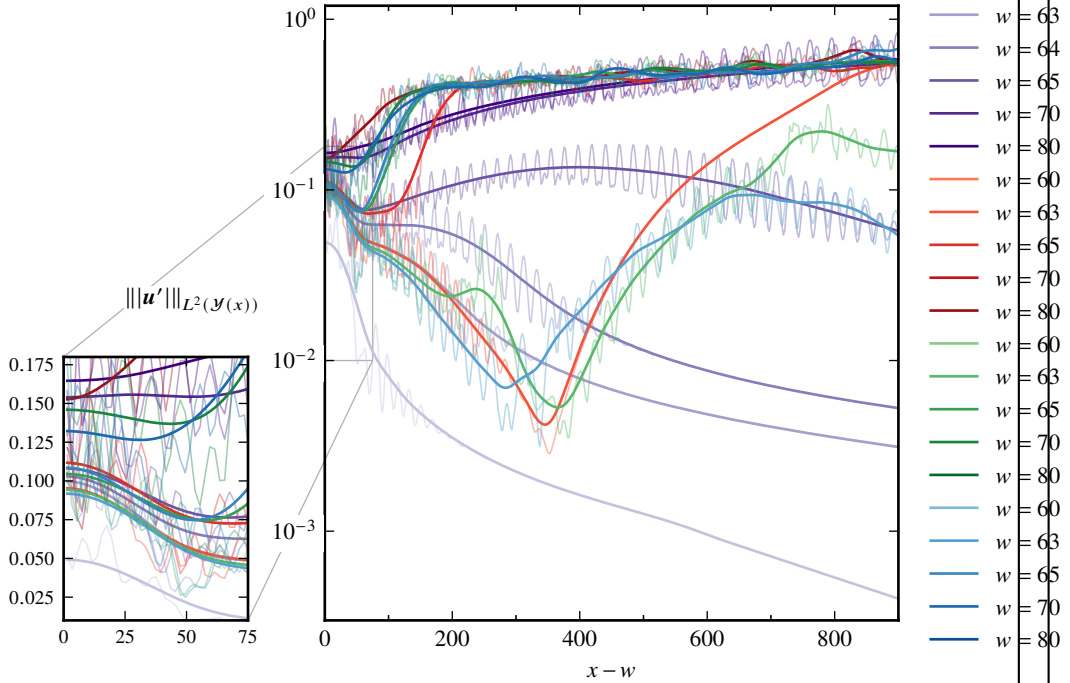


Figure 19. Streamwise distribution of the wall-normal L^2 norm of the magnitude of the perturbation velocity, $|||u' |||_{L^2(y(x))}$, for different gap configurations of variable widths at fixed gap depth $d = 1.5$ and $Re = 1000$. Temporal averaging is performed to obtain the smooth curves, while the low-opacity curves represent an instantaneous snapshot of the trajectories.

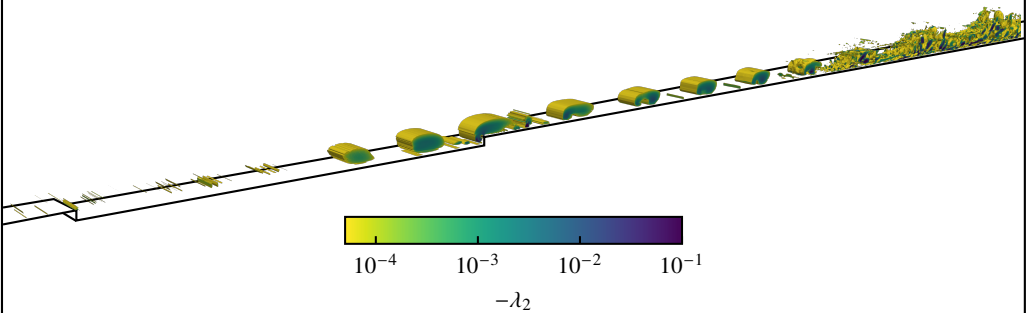


Figure 20. Transition scenario for a gap configuration with $w = 80$ and $d = 1.5$ at $Re = 1000$. Vortical structures are visualised using λ_2 -criterion isosurfaces.

Figure 20 shows a representative shallow-gap case from region IV. Near the downstream edge of the gap the flow displays the two-dimensional oscillation of the underlying periodic orbit. As these structures are advected downstream their frequency falls within the convectively unstable band, so that they are amplified until they reach an amplitude sufficient for breakdown. Transition therefore takes place in the boundary layer well

downstream of the gap, through the primary TS-wave mechanism. The scenario is in this respect distinct from the deep-gap case of figure 17, in which the two-dimensional oscillation is itself unstable to three-dimensional perturbations and the flow breaks down at the gap.

6. Summary and discussion

We have studied how a spanwise-uniform rectangular gap affects the stability and transition of a laminar incompressible boundary layer, using DNS and global stability analysis over the range $w \in [0, 130]$, $d \in [0, 4]$ and $Re \in [800, 3000]$. We have studied the gap from two complementary viewpoints. Where the equilibrium is stable, the gap has been treated as an amplifier of incoming disturbances and characterised by the amplification increment Δn . Beyond that regime, it has been treated as a dynamical system, and the bifurcations through which it loses stability have been followed. Three results stand out.

First, the gap equilibrium loses stability through a supercritical Hopf bifurcation, and the resulting neutral curve is not monotonic in width: increasing the width, the critical depth first decreases, reaching a minimum where the two gap edges interfere most strongly, and then increases towards the isolated-step limit. Increasing the Reynolds number moves the whole boundary towards smaller gaps.

Second, while the equilibrium remains stable the gap acts only as an amplifier to perturbations that reach it from upstream, and this effect is fully captured by the increment Δn obtained from stochastically forced linearised simulations. The resulting Δn contour map shows good agreement with the ΔN experimental values measured by Crouch *et al.* (2022). Local LSA has proven opposite Reynolds-number trends followed by narrow and wide gaps. Increasing Re narrows the band of convectively unstable frequencies but strengthens the shear layer, and the width of the gap decides which of the two prevails. In a narrow gap the higher growth rates act over too short a distance to overcome the loss of bandwidth, so that Δn decreases with Re . In a wide gap the higher growth rates act over a distance long enough to do so, and Δn increases with Re .

Third, transition at the gap is not set by the linear instability of the steady flow. The purely three-dimensional mode of the equilibrium saturates and leaves the downstream boundary layer laminar; breakdown instead requires a two-dimensional limit cycle that is itself unstable to three-dimensional perturbations. This condition is both necessary and sufficient, and the upper envelope of the two- and three-dimensional neutral curves reproduces the experimental bypass boundary of Crouch *et al.* (2022) over the whole range considered.

These results separate two roles that a gap can play, and the distinction determines how transition should be predicted. As long as the flow over the gap remains steady, the gap merely modifies the amplification of a convective instability that it does not create; its whole effect is then contained in the single increment Δn , and the variable- N method (Crouch *et al.* 2006; Crouch 2022) applies. Once the gap sustains a limit cycle that is unstable to three-dimensional perturbations, it generates the disturbances that break the flow down itself. Transition, then, no longer proceeds through the amplification of incoming waves, and no critical N -factor can describe it. What separates the two regimes is a bifurcation of the flow, which can be computed directly and without calibration, and which moves towards smaller gaps as the Reynolds number increases, so that tolerances set at one flight condition do not carry over to another.

REFERENCES

AGUIRRE, FELIPE O., VICTORINO, VICTOR B., MATHIAS, MARLON S. & MEDEIROS, MARCELLO A. F. 2024

- Analysis of the instability of two-dimensional modes in gaps. In *30th AIAA/CEAS Aeroacoustics Conference (2024)*. AIAA Paper 2024-3238.
- BARKLEY, D., BLACKBURN, H. M. & SHERWIN, S. J. 2008 Direct optimal growth analysis for timesteppers. *Int. J. Numer. Meth. Fluids* **57** (9), 1435–1458.
- BEGUET, S., PERRAUD, J., FORTE, M. & BRAZIER, J.-PH. 2017 Modeling of transverse gaps effects on boundary-layer transition. *J. Aircr.* **54** (2), 794–801.
- BRIDGES, THOMAS J. & MORRIS, PHILIP J. 1984 Differential eigenvalue problems in which the parameter appears nonlinearly. *J. Comput. Phys.* **55** (3), 437–460.
- BRÈS, GUILLAUME A. & COLONIUS, TIM 2008 Three-dimensional instabilities in compressible flow over open cavities. *J. Fluid Mech.* **599**, 309–339.
- CHOMAZ, JEAN-MARC 2005 Global instabilities in spatially developing flows: non-normality and nonlinearity. *Annu. Rev. Fluid Mech.* **37**, 357–392.
- CITRO, V., GIANNETTI, F., BRANDT, L. & LUCHINI, P. 2015 Linear three-dimensional global and asymptotic stability analysis of incompressible open cavity flow. *J. Fluid Mech.* **768**, 113–140.
- CROOK, S. D., LAU, T. C. W. & KELSO, R. M. 2013 Three-dimensional flow within shallow, narrow cavities. *J. Fluid Mech.* **735**, 587–612.
- CROUCH, J. D. 2022 Predicting laminar-turbulent transition influenced by surface-induced flow distortions. In *IUTAM Laminar-Turbulent Transition* (ed. Spencer Sherwin, Peter Schmid & Xuesong Wu), *IUTAM Bookseries*, vol. 38, pp. 19–32. Cham: Springer International Publishing.
- CROUCH, J. D., KOSORYGIN, V. S. & NG, L. L. 2006 Modeling the effects of steps on boundary-layer transition. In *6th IUTAM Symposium on Laminar-Turbulent Transition* (ed. R. Govindarajan), *Fluid Mechanics and Its Applications*, vol. 78, pp. 37–44. Dordrecht: Springer.
- CROUCH, J. D., KOSORYGIN, V. S., SUTANTO, M. I. & MILLER, G. D. 2022 Characterizing surface-gap effects on boundary-layer transition dominated by Tollmien–Schlichting instability. *Flow* **2**, E8.
- CROUCH, J. D., NG, L. L., KACHANOV, Y. S., BORODULIN, V. I. & IVANOV, A. V. 2015 Influence of surface roughness and free-stream turbulence on cross-flow-instability transition. In *Procedia IUTAM*, , vol. 14, pp. 295–302. Elsevier.
- DONG, S., KARNIADAKIS, G. E. & CHRYSOSTOMIDIS, C. 2014 A robust and accurate outflow boundary condition for incompressible flow simulations on severely-truncated unbounded domains. *J. Comput. Phys.* **261**, 83–105.
- EHRENSTEIN, UWE & GALLAIRE, FRANÇOIS 2005 On two-dimensional temporal modes in spatially evolving open flows: the flat-plate boundary layer. *J. Fluid Mech.* **536**, 209–218.
- GATICA, GABRIEL N. 2014 *A Simple Introduction to the Mixed Finite Element Method: Theory and Applications*. *Springer Briefs in Mathematics* 1. Springer.
- GENTILI, L. 2012 Experimental study of the effect of gaps on the Laminar/Turbulent transition for incompressible boundary layers. *Tech. Rep.*. ONERA, Toulouse, France, oNERA Internship Report.
- GHARIB, M. & ROSHKO, A. 1987 The effect of flow oscillations on cavity drag. *J. Fluid Mech.* **177**, 501–530.
- HUERRE, PATRICK & MONKEWITZ, PETER A. 1990 Local and global instabilities in spatially developing flows. *Annu. Rev. Fluid Mech.* **22**, 473–537.
- JORDI, BASTIEN E., COTTER, COLIN J. & SHERWIN, SPENCER J. 2014 Encapsulated formulation of the selective frequency damping method. *Phys. Fluids* **26** (3), 034101.
- KARNIADAKIS, GEORGE & SHERWIN, SPENCER 2005 *Spectral/hp Element Methods for Computational Fluid Dynamics*. Oxford University Press.
- KARNIADAKIS, GEORGE EM, ISRAELI, MOSHE & ORSZAG, STEVEN A 1991 High-order splitting methods for the incompressible navier-stokes equations. *J. Comput. Phys.* **97** (2), 414–443.
- KIRBY, ROBERT M. & SHERWIN, SPENCER J. 2006 Stabilisation of spectral/hp element methods through spectral vanishing viscosity: Application to fluid mechanics modelling. *Comput. Methods Appl. Mech. Eng.* **195** (23), 3128–3144, incompressible CFD.
- MALIK, MUJEEB R. 1990 Numerical methods for hypersonic boundary layer stability. *J. Comput. Phys.* **86** (2), 376–413.
- MATHIAS, MARLON S. & MEDEIROS, MARCELLO A. F. 2019 Global instability analysis of a boundary layer flow over a small cavity. In *AIAA Aviation 2019 Forum*. AIAA Paper 2019-3535.
- MATHIAS, MARLON S. & MEDEIROS, MARCELLO A. F. 2021 The effect of incoming boundary layer thickness and Mach number on linear and nonlinear Rossiter modes in open cavity flows. *Theor. Comput. Fluid Dyn.* **35** (4), 495–513.
- MATSUBARA, MASAHARU & ALFREDSSON, P. HENRIK 2001 Disturbance growth in boundary layers subjected to free-stream turbulence. *J. Fluid Mech.* **430**, 149–168.
- MENGALDO, G., DE GRAZIA, D., MOXEY, D., VINCENT, P. E. & SHERWIN, S. J. 2015 Dealiasing techniques for

- high-order spectral element methods on regular and irregular grids. *Journal of Computational Physics* **299**, 56–81.
- MESEGUER-GARRIDO, F., DE VICENTE, J., VALERO, E. & THEOFILIS, V. 2014 On linear instability mechanisms in incompressible open cavity flow. *J. Fluid Mech.* **752**, 219–236.
- MOXEY, DAVID, CANTWELL, CHRIS, BAO, YAN, CASSINELLI, ANDREA, CASTIGLIONI, GIACOMO, CHUN, SEHUN, JUDA, EMILIA, KAZEMI, EHSAN, LACKHOVE, KILIAN, MARCON, JULIAN, MENGALDO, GIANMARCO, SERSON, DOUGLAS, TURNER, MICHAEL, XU, HUI, PEIRO, JOAQUIM, KIRBY, ROBERT & SHERWIN, SPENCER 2020 Nektar++: Enhancing the capability and application of high-fidelity spectral/hp element methods. *Comput. Phys. Commun.* **249**, 107110.
- NENNI, J. P. & GLUYAS, G. L. 1966 Aerodynamic design and analysis of an LFC surface. *Astronaut. Aeronaut.* **4** (7), 52–57.
- OLIVE, M. & BLANCHARD, A. 1982 Étude expérimentale du déclenchement de la transition par des cavités en écoulement incompressible. *Tech. Rep.* TR RT-18/5007. ONERA, Toulouse, France.
- OLSHANSKII, MAXIM A., REUSKEN, ARNOLD & ZHILIAKOV, ALEXANDER 2021 Inf-sup stability of the trace P2–P1 Taylor–Hood elements for surface PDEs. *Math. Comput.* **90** (330), 1527–1555.
- PERRAUD, JEAN, ARNAL, DANIEL & KUEHN, WINFRIED 2014 Laminar-turbulent transition prediction in the presence of surface imperfections. *Int. J. Eng. Syst. Model. Simul.* **6** (3/4), 162–170.
- ROSSITER, J. E. 1964 Wind-tunnel experiments on the flow over rectangular cavities at subsonic and transonic speeds. *Tech. Rep.* R&M No. 3438. Aeronautical Research Council, London.
- ROWLEY, CLARENCE W. & WILLIAMS, DAVID R. 2006 Dynamics and control of high-reynolds-number flow over open cavities. *Annu. Rev. Fluid Mech.* **38**, 251–276.
- SCHMID, PETER J. & HENNINGSON, DAN S. 2001 *Stability and Transition in Shear Flows*, *Applied Mathematical Sciences*, vol. 142. Springer.
- SÉRAUDIE, A. 2010 Influence of gaps and holes on boundary layer, in 2D and 3D flows. *Tech. Rep.* TR RT-4/14790. ONERA, Toulouse, France.
- SERSON, D., MENEGHINI, J. R. & SHERWIN, S. J. 2016 Velocity-correction schemes for the incompressible Navier–Stokes equations in general coordinate systems. *J. Comput. Phys.* **316**, 243–254.
- SINHA, S. N., GUPTA, A. K. & OBERAI, M. M. 1982 Laminar separating flow over backsteps and cavities part II: Cavities. *AIAA J.* **20** (3), 370–375.
- SMITH, A. M. O. & GAMBERONI, N. 1956 *Transition, Pressure Gradient and Stability Theory*. El Segundo Division.
- THEOFILIS, VASSILIOS 2011 Global linear instability. *Annu. Rev. Fluid Mech.* **43**, 319–352.
- VAN INGEN, J. L. 1956 A suggested semi-empirical method for the calculation of the boundary layer transition region. Report. Delft University of Technology, accessed 22/07/2025.
- DE VICENTE, J., BASLEY, J., MESEGUER-GARRIDO, F., SORIA, J. & THEOFILIS, V. 2014 Three-dimensional instabilities over a rectangular open cavity: from linear stability analysis to experimentation. *J. Fluid Mech.* **748**, 189–220.
- ZAHN, JOHANNES & RIST, ULRICH 2016 Impact of deep gaps on Laminar–Turbulent transition in compressible boundary-layer flow. *AIAA J.* **54** (1), 66–76.
- ZHANG, J., MORISHITA, E., OKUNUKI, T. & ITOH, H. 2005 Experimental investigation on the mechanism of flow-type changes in supersonic cavity flows. *Trans. Jpn. Soc. Aeronaut. Space Sci.* **45**, 170–177.